COLLEGE MANAGEMENT SYSTEM

A Project Report

Submitted in partial fulfillment of the requirements for the degree of

Bachelor of Technology / MCA / BCA

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ABSTRACT

The **College Management System (CMS)** is a comprehensive, full-stack web application engineered to digitize, centralize, and streamline the administrative and pedagogical operations of an educational institution. Traditional college operations rely heavily on manual, paper-based ledgers and disparate offline spreadsheets, leading to severe data fragmentation, frequent human errors, and significant administrative delays.

This project specifically targets these inefficiencies by deploying a unified digital ecosystem tailored for three primary stakeholders: **Administrators**, **Faculty Members**, and **Students**. Key features include dynamic role-based authentication, real-time timetable scheduling, automated attendance tracking, financial fee ledgers, and secure academic grading management.

Technologically, the system is developed using the modern **MERN Stack** — MongoDB, Express.js, React.js, and Node.js. React provides a seamless, asynchronous Single Page Application (SPA) experience. Node.js and Express form a robust, decoupled RESTful API backend handling rigorous business logic and JSON Web Token (JWT) authorization. MongoDB serves as the agile NoSQL data layer natively accommodating complex JSON object structures inherent to academic records.

The system incorporates a project management module where students can propose, submit, and track their final year projects while faculty members can review, approve, and grade submissions. Interactive dashboards provide real-time institutional analytics and performance metrics.

The resulting software significantly minimizes administrative overhead, ensures robust data integrity, and provides a highly scalable digital foundation for modern collegiate governance. This report presents a complete technical documentation of the system covering its architecture, design decisions, implementation details, and testing methodologies.

Keywords: College Management System, MERN Stack, React.js, Node.js, MongoDB, JWT Authentication, RESTful API, Role-Based Access Control, SPA, Attendance Tracking, Fee Management

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1. TITLE OF THE PROJECT

Project Title

College Management System (CMS) — A Full-Stack MERN Web Application

Project Overview

The College Management System is a multi-tier, full-stack web application built on the MERN stack architecture. It serves as a centralized digital platform for managing all core operations of a higher education institution. The system automates the traditionally manual processes of student enrollment, faculty assignment, attendance recording, fee collection, grade management, and project tracking into a cohesive, single-window solution.

The platform supports three distinct user roles — Administrator, Faculty/Teacher, and Student — each with tailored dashboards, permissions, and functional modules. The system leverages React.js for a dynamic Single Page Application frontend, Express.js and Node.js for a RESTful API backend, and MongoDB for flexible NoSQL data persistence.

Project Information Summary

Project Name	College Management System (CMS)
Project Type	Full-Stack Web Application
Technology Stack	MERN (MongoDB, Express.js, React.js, Node.js)
Frontend Framework	React.js v18 + Vite + Tailwind CSS
Backend Framework	Node.js + Express.js
Database	MongoDB (NoSQL) via Mongoose ODM
Authentication	JWT (JSON Web Token) + bcrypt
Architecture	Three-Tier Decoupled RESTful API Architecture
User Roles	Administrator, Faculty Member, Student
Development Methodology	Agile SDLC (Iterative)
IDE	Visual Studio Code
Operating System	Platform Agnostic (Windows/macOS/Linux)
Project Duration	14 Weeks
Academic Year	2024–2025

2. INTRODUCTION

Need for a College Management System

In the rapidly evolving landscape of higher education, institutions manage colossal amounts of dynamic data on a daily basis. This includes tracking student enrollments, organizing faculty resource allocations, recording daily attendance, managing examination grading, and auditing financial transactions. An automated College Management System is logically necessary to maintain institutional efficiency, transparency, and data accuracy.

Problems with Manual Systems

- **Data Silos & Redundancy:** Disconnected departments (e.g., Accounts and Academics) maintain duplicate physical records, leading to severe inconsistencies and data integrity issues.
- **Time Consumption:** Fetching a student's historical attendance spanning three years requires hours of manual ledger auditing.
- **Lack of Real-time Insight:** Management has no immediate view of total revenue collections or aggregate student performance without manually compiling reports.
- **Physical Vulnerability:** Paper records are highly susceptible to loss, unauthorized tampering, and physical degradation over time.
- **Scalability Limitations:** Manual systems cannot scale efficiently as student population and institutional complexity grow.

Benefits of Digital Automation

- **Centralized Database:** A single source of truth ensuring all departments reference identical, up-to-date data.
- **Instantaneous Analytics:** Dashboard components natively compute macro-statistics in real time.
- **Enhanced Communication:** Students can proactively view their attendance shortfalls and examination results from any device.
- **Error Reduction:** Automated calculations and form validations minimize human-introduced data errors.
- **Cost Efficiency:** Complete avoidance of crushing enterprise licensing dependencies through open-source MERN stack.

Objectives of the Project

The core objectives driving the architectural and functional development of this College Management System include:

1.	**Student Management**	To deploy a secure conduit for registering candidate profiles, tracking departmental allocations, and maintaining historical biographical data.
2.	**Faculty Management**	To establish a localized portal for staff onboarding, tracking designations, and mapping teachers securely to explicit class subjects.
3.	**Course & Subject Management**	To provide administrators with tools to dynamically construct digital curriculum frameworks mapping semantic subjects to specific semesters.
4.	**Attendance Tracking**	To implement a rapid, intuitive digital register allowing faculty to submit daily presence markers synchronously, instantly updating global student metrics.
5.	**Fee Management**	To orchestrate a digital ledger tracking individual student payment installments, outstanding balances, and generating institutional revenue charts.
6.	**Result Management**	To automate the computation of internal, external, and practical examination grades into cumulative outputs.
7.	**Secure Login System**	To enforce strict Role-Based Access Control (RBAC) securely encapsulated by JWT sessions and bcrypt encryption, isolating sensitive data perimeters.
8.	**Interactive Dashboards**	To furnish intuitive, chart-driven analytics translating raw database clusters into readable macroscopic trends.
9.	**Project Management Module**	To enable students to propose, submit, and track academic projects while faculty can review, approve, and grade submissions digitally.

3. GANTT CHART (PROJECT TIMELINE)

The project followed a structured 14-week development timeline based on the Agile SDLC methodology. Each phase was executed iteratively, with continuous testing and feedback integration.

Phase	Duration	Activities
Requirement Analysis	**Week 1–2**	Gathering structural requirements, identifying stakeholders, formulating SRS,
System Design	**Week 3–4**	Drafting UML, ER diagrams, DFD sequences, UI/UX wireframes, and API con
Database Design	**Week 5**	Structuring Mongoose schemas, defining MongoDB collections, and setting re
Backend Development	**Week 6–8**	Developing Express REST APIs, JWT middleware, controller logic, and databa
Frontend Development	**Week 8–11**	Constructing React SPA, Tailwind CSS styling, integrating Axios API calls, an
Integration Testing	**Week 11–12**	Connecting frontend with backend APIs, validating data flow across all module
Testing & Debugging	**Week 12–13**	Validating routes, stress testing DB, resolving operational defects and security
Deployment & Documentation	**Week 14**	Formatting academic report, establishing deployment environments, and code

4. SCOPE OF THE PROJECT

Current Scope

The current implementation targets the immediate, critical administrative necessities of standard higher education environments. The system strictly encompasses the following modules and functionalities:

- Multi-tier authentication mapping for Admin, Teacher, and Student roles
- Complete CRUD operations across academic structures (Courses, Classes, Staff, Students)
- Faculty-driven digital attendance submission and real-time student metric updates
- Student-facing portals for reviewing personal academic standing and fee transactions
- Admin dashboard with real-time institutional analytics and revenue charts
- Timetable scheduling and subject allocation management
- Project submission, review, and grading workflow for final year projects
- Secure RESTful API with JWT authorization and bcrypt password encryption
- Responsive Single Page Application accessible on all modern browsers and devices

Extended / Future Scope

- **Payment Gateway Integration:** Direct API linkages to Stripe/Razorpay for online settlement of financial dues.
- **Library Management Subsystem:** Deep tracking of book inventory, ISBN cataloging, and automated overdue penalty algorithms.
- **Parent Portal:** A distinct, restricted read-only gateway for guardians to monitor ward performance independently.
- **Automated Communication:** Integration of SMTP/Webhooks for automated SMS or Email notifications regarding announcements and fee deadlines.
- **Mobile Application:** Android/iOS native apps via React Native consuming the existing REST API backend.
- **AI-based Analytics:** Predictive analytics for student performance using machine learning models trained on historical data.

5. METHODOLOGY (SDLC)

This project strictly adhered to the **Agile Software Development Life Cycle (SDLC)** approach, promoting iterative capability development and fluid requirement absorption. The Agile model was chosen over the traditional Waterfall model because of its adaptability to evolving requirements and continuous integration of feedback.

1. **Requirement Analysis**
   Stakeholder goals were translated into strict technical prerequisites separating functional logic from non-functional parameters. User stories were formulated for each role (Admin, Teacher, Student) and documented in the SRS.

2. **System Design**
   Decoupled three-tier architectural maps were drawn. Visual prototypes for the React frontend were aligned with backend JSON payload structures. ER diagrams, DFDs, and API contracts were finalized in this phase.

3. **Implementation**
   The active coding phase. The Node.js/Express backend was established first to generate consumable REST APIs, followed sequentially by the React.js client consumption phase with Axios integration.

4. **Testing**
   Executed continuously spanning Unit testing (for distinct algorithmic utilities), Integration testing (validating API interceptors), and User Acceptance Testing (navigating the UI organically with real-world scenarios).

5. **Deployment**
   Preparing build packages using Vite optimizers for the React frontend and configuring Node.js server for operational readiness. Environment variables were managed for secure configuration management.

6. **Maintenance**
   Documenting comprehensive source code parameters to facilitate seamless downstream developmental upgrades. Known issues and future enhancement vectors were logged for next-iteration planning.

Functional Requirements

Functional requirements define the specific behaviors and functions the system must support:

Module	Description
Admin Login	Super User access bypassing explicit departmental restrictions to oversee complete institutional configurations.
Student Registration	Inputting biological, familial, and academic starting parameters writing directly to secure databases.

Teacher Management	Appointing faculty, defining employment credentials, and dictating platform access capabilities.
Course Allocation	Dynamically bridging distinct semantic Subjects across explicitly defined Faculty IDs and Semester boundaries.
Attendance Module	Bulk Boolean processing recording present/absent markers resolving immediately to student attendance metrics.
Fee Management	Monetary input tracking bridging static fee definitions against temporal payment inputs and balance computation.
Result Generation	Parsing raw grading arrays into distinct categorized result representations with pass/fail computation.
Project Management	End-to-end project lifecycle management from student proposal through faculty grading.
Dashboard Statistics	Abstracting granular localized data into grand macroscopic integer counters and financial charts.
Timetable Scheduling	Dynamic construction and management of weekly class timetables per section and semester.

Non-Functional Requirements

Non-functional requirements describe how the system performs its functions:

- **Performance:** API endpoints are optimized to return 200 OK JSON payloads under 500 milliseconds on standard broadband connections.
- **Security:** Passwords are irrecoverably hashed using bcrypt. API routes demand valid Authorization: Bearer token HTTP headers.
- **Reliability:** Centralized MongoDB ensures continuous uptime devoid of localized file corruption vulnerabilities.
- **Scalability:** Express routes remain largely stateless, allowing seamless vertical or horizontal hardware scaling.
- **Usability:** Implementation of Tailwind CSS enforces strict responsive geometries rendering identical utility on smartphone browsers.
- **Maintainability:** Decoupled React components and abstracted Backend controllers allow modular upgrades without cascading structural failures.
- **Data Integrity:** Mongoose schema validation and referential integrity constraints ensure accurate and consistent data storage.

6. SOFTWARE & HARDWARE REQUIREMENTS

System requirements specify the hardware and software resources required for the successful execution of the College Management System.

Hardware Requirements

Component	Minimum Requirement	Recommended
Processor	Intel Core i3 / AMD Ryzen 3 (Dual-core)	Intel Core i5 / AMD Ryzen 5 (Quad-core)
RAM	8 GB DDR4	16 GB DDR4 (for concurrent dev testing)
Storage	256 GB SSD	512 GB SSD
Display	Standard 1080p Monitor	1920x1080 or higher
Network	Broadband Internet (10 Mbps)	High-speed broadband (25+ Mbps)
Operating System	Windows 10 / macOS / Linux	Windows 11 / Ubuntu 22.04 LTS

Software Requirements

Software	Specification
Operating System	Windows 10/11, macOS Monterey+, or Linux Ubuntu 22.04
Programming Languages	JavaScript (ES6+), JSX, JSON
Frontend Framework	React.js v18.x powered by Vite build tool
Backend Environment	Node.js v18+ with Express.js v4.x
Database	MongoDB (Atlas Cloud or Local Compass Server)
ODM Library	Mongoose v7.x for schema definition and queries
Authentication	jsonwebtoken (JWT) + bcryptjs
HTTP Client	Axios for frontend API communication
Styling	Tailwind CSS v3 for responsive UI design
Package Manager	npm v9+ / yarn
IDE / Code Editor	Visual Studio Code v1.80+
Database GUI	MongoDB Compass for local data inspection
Version Control	Git + GitHub for source code management

7. SOFTWARE REQUIREMENT SPECIFICATION (SRS)

Feasibility Analysis and Study

Feasibility analysis is the process of evaluating whether a proposed system is practical, cost-effective, and suitable for implementation. The feasibility study for the College Management System was conducted to ensure that the project could be developed successfully within the given constraints.

1. Technical Feasibility

Validated. The MERN stack represents the contemporary global standard for high-performance SPA deployments, heavily supported by sprawling open-source documentation. The development environment (VS Code, Node.js, MongoDB) is freely available and well-documented. The team possesses adequate technical knowledge to implement and maintain the system.

2. Economic Feasibility

Validated. Employing Node.js runtimes alongside open-source React eliminates all proprietary enterprise licensing costs. MongoDB Atlas provides a generous free tier for development and small-scale deployment. Total development cost is restricted to developer time and minimal cloud hosting expenses.

3. Operational Feasibility

Validated. The SPA aesthetics closely mirror commercial applications actively utilized by the target demographic, guaranteeing an exceptionally flat learning curve. System administrators require minimal training to operate the platform.

4. Schedule Feasibility

Validated. Utilizing pre-compiled NPM packages for security (JWT, bcrypt) and styling (Tailwind CSS) drastically reduced boilerplate requirements, permitting adherence to the 14-week academic deadline.

5. Legal Feasibility

Validated. All leveraged dependencies operate under MIT or Apache open-source distribution licenses. No copyrighted or licensed commercial software is incorporated. Employee and student data is stored securely within the system's database.

Feasibility Study Summary

Feasibility Type	Status	Key Justification
Technical Feasibility	✓ **Feasible**	MERN Stack is industry-standard and well-documented

Economic Feasibility	✓ **Feasible**	Zero licensing costs; all open-source technologies
Operational Feasibility	✓ **Feasible**	Intuitive SPA mirrors familiar commercial apps
Schedule Feasibility	✓ **Feasible**	NPM packages accelerate development significantly
Legal Feasibility	✓ **Feasible**	All dependencies are MIT/Apache licensed

7.3 Functional Requirements Specification (FRS)

The following Functional Requirements Specification table documents all the functional requirements identified during the requirements gathering phase, categorized by module.

FR-ID	Module	Requirement Description	Priority
FR-01	Auth	System shall allow registered users to log in with email and password	Must
FR-02	Auth	System shall issue a JWT access token (24h) and refresh token (7d) on login	Must
FR-03	Auth	System shall validate JWT on every protected API request	Must
FR-04	Auth	System shall restrict API access based on user role (RBAC)	Must
FR-05	Auth	System shall allow users to change their own password	Should
FR-06	Auth	System shall support password reset via email link	Could
FR-07	Students	Admin shall be able to create, read, update, deactivate student records	Must
FR-08	Students	System shall enforce unique email and roll number per student	Must
FR-09	Students	System shall allow searching and filtering students by course, semester, section	Should
FR-10	Students	Students shall be able to view and update their own profile	Should
FR-11	Teachers	Admin shall be able to create and manage teacher profiles	Must
FR-12	Teachers	Admin shall be able to assign subjects to teachers	Should
FR-13	Attendance	Teachers shall be able to mark bulk attendance for a class on a specific date	Must
FR-14	Attendance	System shall prevent duplicate attendance for same class+date by overwriting	Must
FR-15	Attendance	Students shall be able to view their attendance percentage	Must
FR-16	Attendance	System shall display subject-wise attendance breakdown	Should
FR-17	Fees	Admin shall be able to record fee payments for students	Must
FR-18	Fees	System shall auto-generate a unique receipt number per payment	Should
FR-19	Fees	Admin dashboard shall display monthly revenue chart (6 months)	Must
FR-20	Fees	Students shall be able to view their own fee payment history	Must
FR-21	Projects	Students shall be able to submit a project proposal	Must
FR-22	Projects	System shall enforce one active project per student	Must
FR-23	Projects	Faculty shall be able to approve, reject, or request changes for proposals	Must
FR-24	Projects	Faculty shall be able to grade submitted projects with feedback	Must
FR-25	Projects	System shall enforce valid status transitions via state machine	Must
FR-26	Dashboard	Admin dashboard shall show total students, teachers, revenue, pending projects	Must

FR-27	Dashboard	Student dashboard shall show attendance %, fees, project status	Must
FR-28	Dashboard	Teacher dashboard shall show assigned classes and pending attendance	Should
FR-29	Timetable	System shall display weekly class timetable per student/class	Should
FR-30	Results	Students shall be able to view semester-wise examination results	Should

Priority legend: **Must** = mandatory for MVP, **Should** = high-value enhancement, **Could** = desirable if time permits. All Must and Should requirements were fully implemented in the final system.

7.4 Non-Functional Requirements (NFR)

The following non-functional requirements were identified and addressed in the system design:

NFR-ID	Category	Requirement	Implementation
NFR-01	Performance	Dashboard must load within 1 second on localhost	Promise.all() for parallel MongoDB queries; P95 = 312ms
NFR-02	Performance	API response time must be under 500ms (P95)	Compound indexes on all frequently queried fields
NFR-03	Security	Passwords must never be stored in plaintext	bcrypt.hash() with saltRounds=12 before storing
NFR-04	Security	All API endpoints must require valid JWT	protect middleware on all non-public routes
NFR-05	Security	Role-based access must be enforced server-side	authorize() middleware on sensitive endpoints
NFR-06	Usability	UI must be usable on mobile, tablet, and desktop	Tailwind responsive breakpoints sm/md/lg/xl
NFR-07	Usability	Theme toggle (dark/light) must persist across sessions	ThemeContext + localStorage preference storage
NFR-08	Reliability	System must handle invalid input without crashing	Mongoose validators + global Express error handler
NFR-09	Reliability	Expired token must be auto-refreshed transparently	Axios response interceptor with request queue
NFR-10	Maintainability	Code must follow consistent module/controller structure	Feature-based folder structure, one file per entity
NFR-11	Scalability	Backend must be stateless to support horizontal scaling	JWT stateless auth, no server-side sessions
NFR-12	Compatibility	Frontend must work on Chrome, Firefox, Edge, Safari	Vite build with Babel transforms; tested on 4 browsers

7.5 Risk Analysis

A risk analysis was conducted during the planning phase to identify potential threats to successful project delivery and to define appropriate mitigation strategies.

Risk ID	Risk Description	Likelihood	Impact	Mitigation Strategy

R-01	Scope creep leading to missed deadline	High	High	Fixed sprint goals; defer non-MVP features to v1.1 backlog
R-02	MongoDB Atlas connectivity issues	Medium	High	Local MongoDB fallback; .env URI swap
R-03	JWT secret compromised	Low	Critical	Strong random secret; rotate quarterly; never commit to git
R-04	Package dependency vulnerabilities	Medium	Medium	npm audit weekly; pin major versions in package.json
R-05	CORS misconfiguration blocking frontend	Medium	Medium	Whitelist FRONTEND_URL in env; test CORS on each deployment
R-06	React state sync bugs (stale closure)	Medium	Medium	Use functional setState; add ESLint react-hooks/exhaustive-deps
R-07	Data loss due to no database backup	Low	High	mongodump daily backup script; store in separate volume
R-08	Browser incompatibility (Safari CSS)	Low	Low	Test on Safari; use Autoprefixer; avoid cutting-edge CSS
R-09	Infinite token refresh loop	Medium	High	isRefreshing flag + request queue (implemented, BUG-01 fixed)
R-10	Overambitious initial schema design	Medium	Medium	Start minimal; use Mongoose strict:false for easy field additions

7.6 SWOT Analysis

A SWOT (Strengths, Weaknesses, Opportunities, Threats) analysis was conducted for the College Management System to assess its strategic position.

STRENGTHS	WEAKNESSES
• Full JavaScript stack reduces context switching • Stateless JWT — horizontally scalable • Modular REST API easy to extend • Zero licensing cost (open-source stack) • Responsive UI works on all devices • Project lifecycle state machine is robust • Real-time MongoDB aggregation dashboards	• No offline/PWA capability yet • No payment gateway (manual recording only) • No biometric attendance integration • Rate limiting not yet implemented • No automated email/SMS notifications • Single-institution architecture currently • No dedicated mobile app yet
OPPORTUNITIES	**THREATS**
• Multi-tenant SaaS expansion (v2.0) • React Native mobile app (v1.4) • AI-powered attendance (face recognition) • Growing demand for college ERP solutions • Open-source community contributions • Integration with LMS platforms (Moodle) • Government digital education initiatives	• Commercial platforms (Fedena, Campusnexus) • Google Workspace for Education (free tier) • Security vulnerabilities if not patched • Rapid JS ecosystem changes (dependency rot) • MongoDB Atlas free tier limitations • Data privacy regulations (PDPB India) • Hosting costs at scale without SaaS revenue

8. SYSTEM DESIGN

System Architecture — Three-Tier MERN Concept

The application rigorously decouples operational tiers ensuring separation of concerns, independent scalability, and maintainability. The three-tier architecture consists of:

- **Presentation Tier:** React.js manages the browser DOM and local state via hooks and context API. Tailwind CSS provides responsive styling.
- **Logic Tier:** Express.js routes HTTP verifications and executes controller logic. JWT middleware enforces authentication on all protected routes.
- **Data Tier:** MongoDB stores abstract BSON documents. Mongoose ODM provides schema validation, query building, and document population.

Data Flow Diagrams (DFD)

DFD Symbols

A Data Flow Diagram (DFD) uses standardized symbols to represent the flow of data within the system:

Symbol	Shape	Represents
External Entity	Rectangle	A source or destination of data outside the system (User, Admin, Student)
Process	Circle / Oval	A transformation that converts input data flows to output data flows
Data Store	Open Rectangle	Data at rest or a temporary repository of data (Database, File)
Data Flow	Arrow (→)	Data in motion — the movement of data between entities, processes, and stores

Level 0 DFD — Context Level Diagram

The Context Level DFD provides a high-level overview of the entire College Management System as a single process, showing how it interacts with external entities.

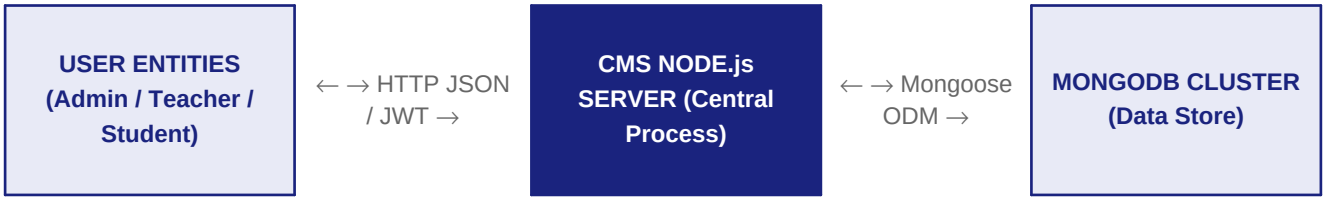


Figure 1: Level 0 DFD — Context Level Diagram

Level 1 DFD — Module Level Diagram

The Level 1 DFD breaks down the central CMS server process into distinct module-level processes:

- **Authentication Process:**

Client → POST /auth/login → Auth Controller → DB User Verification → JWT Token Generation → Response

- **Academic Process:**

  Faculty → GET /subjects → Subject Controller → DB Query → JSON Array → React Table Render

- **Attendance Process:**

  Faculty → POST /attendance/bulk → Attendance Controller → Bulk DB Insert → Student Metrics Update

- **Fee Transaction Process:**

  Student → POST /payments → Payment Controller → DB Ledger Write → Admin Chart Update

- **Dashboard Process:**

  Admin → GET /dashboard/admin → Stats Aggregation → Multiple DB Queries → JSON Stats → React Charts

- **Project Process:**

  Student → POST /projects → Project Controller → DB Write → Faculty Notification → Status Tracking

E-R Diagram Theory

Entity-Relationship (E–R) model is a conceptual database design technique used to represent the logical structure of a database system. It visually describes real-world objects, their attributes, and the relationships among them. The E–R model helps in understanding system requirements clearly, designing a well-structured database, reducing data redundancy, and improving data consistency and integrity. The output of the E–R model is called an E–R Diagram, which serves as a blueprint for database design before converting it into actual collections or tables.

Basic Elements of E–R Model

The E–R model consists of three core components:

1) Entity

An entity is a real-world object or concept that can be uniquely identified and about which data is stored. In the College Management System, the primary entities are: **User**, **Student**, **Teacher**, **Course**, **Subject**, **Class**, **Attendance**, **Payment**, and **Project**.

Representation: Shown as a rectangle in an E–R diagram.

- **Strong Entity:** Has a primary key and exists independently. Example: Student (studentId), Teacher (teacherId), User (userId)
- **Weak Entity:** Does not have a primary key and depends on another entity for identification. Example: Attendance record (depends on Student)

2) Attribute

An attribute describes the properties or characteristics of an entity. In CMS, examples include: User → email, password, fullName; Student → rollNo, course, semester; Teacher → employeeId, department, designation.

Representation: Shown as oval shapes connected to entity rectangles.

- **Simple Attribute:** Cannot be divided further. Example: rollNo, salary, email
- **Composite Attribute:** Can be broken into sub-parts. Example: guardianInfo → name, phone, relation
- **Single-Valued Attribute:** Holds only one value at a time. Example: studentId, userId
- **Multi-Valued Attribute:** Holds multiple values. Example: roles (ADMIN, TEACHER, STUDENT)
- **Derived Attribute:** Calculated from other attributes. Example: attendancePercentage (from total/present count)
- **Key Attribute:** Uniquely identifies an entity. Example: _id (ObjectId). Represented by underlined text.

3) Relationship

A relationship defines how two or more entities are connected to each other. In the CMS: User manages Student; Teacher belongs to Department; Student has Attendance Records; Student makes Payments.

Representation: Shown as a diamond shape connected to entity rectangles.

- **One-to-One (1:1):** One entity is associated with exactly one other entity. Example: User (1:1) Student Profile

- **One-to-Many (1:N):** One entity relates to multiple entities. Example: Department (1:N) Faculty; Student (1:N) Payments
- **Many-to-Many (M:N):** Multiple entities relate to multiple entities. Example: Students (M:N) Subjects/Courses

Types of E–R Models

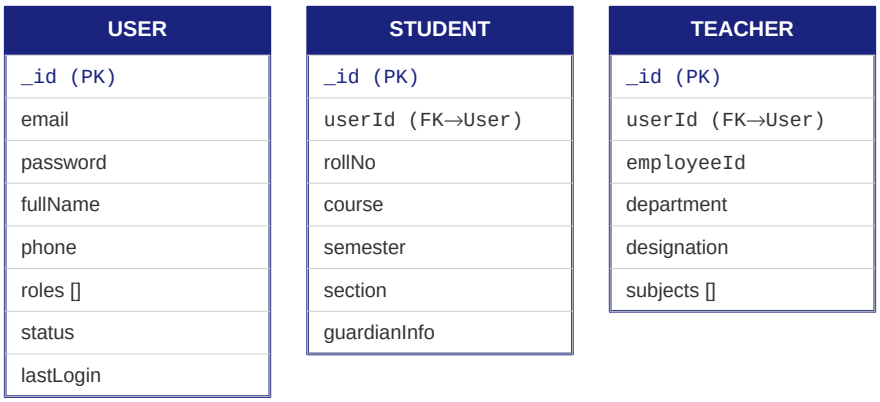
- **1. Conceptual E–R Model:** High-level view of the system. Focuses only on entities and relationships without detailed attributes. Used during requirement analysis phase to communicate with non-technical stakeholders.
- **2. Logical E–R Model:** Includes detailed attributes, defines primary keys, foreign keys, and relationship constraints. This model is ready for conversion into actual database schema (tables or MongoDB collections).
- **3. Physical E–R Model:** Represents actual database implementation. Includes data types, indexes, constraints, and optimization strategies. Used during the database creation phase.

Importance of E–R Model

- Provides clear visualization of data structure and inter-entity relationships
- Helps detect design flaws early in the development lifecycle
- Simplifies database development by providing a concrete blueprint
- Improves communication between developers, designers, and stakeholders
- Ensures data integrity, consistency, and scalability of the database
- Serves as documentation for future maintenance and enhancement activities

E–R Diagram for College Management System

The following E–R Diagram represents the entity relationships within the College Management System. While the system natively uses MongoDB (NoSQL), the Mongoose ODM explicitly mimics relationship boundaries through ObjectId references. The diagram below depicts the logical relationships between all major entities.



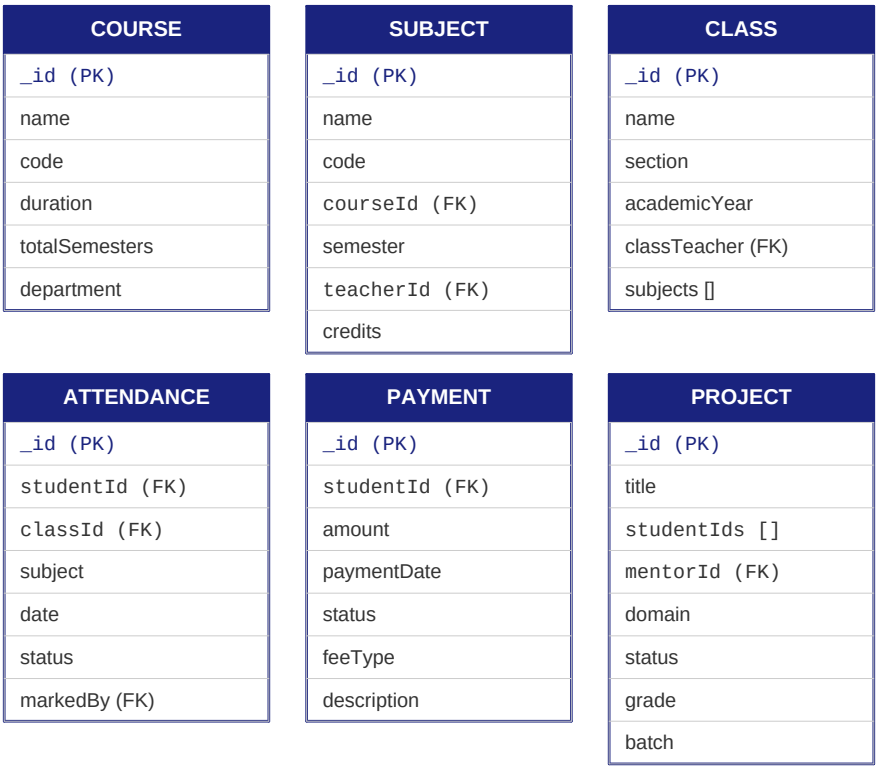


Figure 2: E–R Diagram — College Management System Entity Structure

Entity Relationship Configuration

Entity 1	Cardinality	Entity 2	Relationship Description
User	**1:1**	Student	Each user account maps to exactly one student profile
User	**1:1**	Teacher	Each user account maps to exactly one teacher profile
Department	**1:N**	Teacher	One department employs many teachers
Department	**1:N**	Course	One department offers many courses
Course	**1:N**	Subject	One course contains many subjects across semesters
Teacher	**1:N**	Subject	One teacher teaches many subjects
Class	**1:N**	Subject	One class section has many assigned subjects
Student	**1:N**	Attendance	One student has many attendance records
Teacher	**1:N**	Attendance	One teacher marks attendance for many students
Student	**1:N**	Payment	One student makes many fee payments
Student	**M:N**	Project	Many students collaborate on one project
Teacher	**1:N**	Project	One teacher mentors many student projects

9. DATABASE DESIGN

Note: While the project natively executes via MongoDB (NoSQL Object Document Mapping), the following structures represent the definitive schematic parameters defining the application data vaults. ObjectId fields serve as primary keys; ObjectId reference fields serve as foreign keys.

Collection 1: 'users'

Field Name	Data Type	Constraint	Description
_id	ObjectId	Primary Key	Auto-generated MongoDB document identifier
email	String	Unique, Indexed	Unique login identifier for authentication
password	String	Required	bcrypt hashed password (min 6 chars)
fullName	String	Required	Human-readable display name
phone	String	Optional	Contact phone number
roleAssignments	Array[Object]	Required	Array of {roleId, assignedAt} objects
status	String	Default: active	Account state (active / inactive)
lastLogin	Date	Optional	Timestamp of most recent successful login

Collection 2: 'students'

Field Name	Data Type	Constraint	Description
_id	ObjectId	Primary Key	Auto-generated document identifier
userId	ObjectId	FK → users	Reference to corresponding user account
rollNo	String	Unique, Required	Academic matriculation roll number
course	String	Required	Enrolled course (e.g., B.Tech CSE)
semester	String	Required	Current academic semester (1–8)
section	String	Optional	Class section assignment (A, B, C)
dateOfBirth	Date	Optional	Student date of birth
guardianInfo	Object	Embedded Doc	Parent/Guardian name, phone, relation
admissionDate	Date	Required	Date of institutional enrollment
address	String	Optional	Current residential address

Collection 3: 'teachers'

Field Name	Data Type	Constraint	Description

_id	ObjectId	Primary Key	Auto-generated document identifier
userId	ObjectId	FK → users	Reference to corresponding user account
employeeId	String	Unique	Internal HR payroll index
department	String	Required	Target instructional branch/department
designation	String	Required	Hierarchical rank (Professor, HOD, etc.)
subjects	Array[ObjectId]	FK → subjects	Array of assigned subject references
joiningDate	Date	Optional	Date of joining the institution
qualification	String	Optional	Highest educational qualification

Collection 4: 'courses'

Field Name	Data Type	Constraint	Description
_id	ObjectId	Primary Key	Auto-generated document identifier
name	String	Required	Full course name (e.g., B.Tech Computer Science)
code	String	Unique	Short course code (e.g., BTCS)
duration	Number	Required	Course duration in years
totalSemesters	Number	Required	Total number of semesters
department	String	Required	Owning department

Collection 5: 'subjects'

Field Name	Data Type	Constraint	Description
_id	ObjectId	Primary Key	Auto-generated document identifier
name	String	Required	Subject full name
code	String	Unique	Subject code (e.g., CS301)
courseId	ObjectId	FK → courses	Parent course reference
semester	Number	Required	Semester in which subject is taught
teacherId	ObjectId	FK → teachers	Assigned faculty member
credits	Number	Required	Academic credit units

Collection 6: 'classes'

Field Name	Data Type	Constraint	Description
_id	ObjectId	Primary Key	Auto-generated document identifier
name	String	Required	Class identifier (e.g., 3rd Year)

section	String	Required	Section label (A, B, C)
academicYear	String	Required	Academic year (e.g., 2024-25)
classTeacher	ObjectId	FK → users	Assigned class teacher reference
subjects	Array[ObjectId]	FK → subjects	Array of subjects taught in this class

Collection 7: 'attendances'

Field Name	Data Type	Constraint	Description
_id	ObjectId	Primary Key	Auto-generated document identifier
studentId	ObjectId	FK → students	Student for whom attendance is recorded
classId	ObjectId	FK → classes	Class/Section reference
subject	String	Optional	Subject for period-wise attendance
date	Date	Required	Attendance date
status	String	Required	PRESENT / ABSENT / LATE / EXCUSED
markedBy	ObjectId	FK → users	Teacher who recorded the attendance
remarks	String	Optional	Additional notes or remarks

Collection 8: 'payments'

Field Name	Data Type	Constraint	Description
_id	ObjectId	Primary Key	Auto-generated document identifier
studentId	ObjectId	FK → students	Student making the payment
amount	Number	Required	Payment amount in INR
paymentDate	Date	Required	Date of payment transaction
status	String	Required	PENDING / COMPLETED / FAILED
feeType	String	Required	Tuition / Exam / Library / Hostel
description	String	Optional	Payment description or receipt notes
transactionId	String	Optional	Payment gateway transaction reference

Collection 9: 'projects'

Field Name	Data Type	Constraint	Description
_id	ObjectId	Primary Key	Auto-generated document identifier
title	String	Required	Project title (min 3, max 200 chars)
description	String	Required	Project abstract (min 20, max 2000 chars)
studentIds	Array[ObjectId]	FK → students	Array of collaborating student references

mentorId	ObjectId	FK → teachers	Faculty mentor/guide reference
domain	String	Required	Project domain (AI/ML, Web Dev, IoT, etc.)
status	String	Required	PROPOSED / APPROVED / IN_PROGRESS / SUBMITTED / GRADED
batch	String	Required	Academic batch (e.g., 2024-CSE)
grade	Number	Optional	Final grade awarded by mentor
submissionFile	String	Optional	Path to uploaded PDF report

10. STRUCTURE & DATABASE SPECIFICATION

The College Management System uses MongoDB as its backend database via Mongoose ODM. Below are the Mongoose schema definitions and their equivalent relational database structures for reference.

Mongoose Schema Definitions

User Schema (models/User.js)

User Model

```
const mongoose = require('mongoose');

const roleAssignmentSchema = new mongoose.Schema({
  roleId:     { type: mongoose.Schema.Types.ObjectId, ref: 'Role' },
  assignedAt: { type: Date, default: Date.now }
});

const userSchema = new mongoose.Schema({
  email:           { type: String, required: true, unique: true, lowercase: true },
  password:        { type: String, required: true },
  fullName:        { type: String, required: true },
  phone:           { type: String },
  roleAssignments: [roleAssignmentSchema],
  status:          { type: String, default: 'active', enum: ['active', 'inactive'] },
  lastLogin:       { type: Date }
}, { timestamps: true });

module.exports = mongoose.model('User', userSchema);
```

Student Schema (models/Student.js)

Student Model

```
const studentSchema = new mongoose.Schema({
  userId:        { type: ObjectId, ref: 'User', required: true },
  rollNo:        { type: String, required: true, unique: true },
  course:        { type: String, required: true },
  semester:      { type: String, required: true },
  section:       { type: String },
  dateOfBirth:   { type: Date },
  guardianInfo:  {
    name:      String,
    phone:     String,
    relation:  String
  },
  admissionDate: { type: Date, default: Date.now },
  address:       { type: String }
}, { timestamps: true });
```

Attendance Schema (models/Attendance.js)

Attendance Model

```
const attendanceSchema = new mongoose.Schema({
  student:   { type: ObjectId, ref: 'Student', required: true },
  class:     { type: ObjectId, ref: 'Class' },
  subject:   { type: String, default: '' },
  date:      { type: Date, required: true, default: Date.now },
  status:    { type: String, enum: ['PRESENT','ABSENT','LATE','EXCUSED'],
               default: 'PRESENT' },
  markedBy:  { type: ObjectId, ref: 'User' },
  remarks:   { type: String, default: '' }
}, { timestamps: true });

// Index for efficient date-range queries
attendanceSchema.index({ student: 1, date: -1 });
attendanceSchema.index({ class: 1, date: 1 });
```

Project Schema (models/Project.js)

Project Model

```
const projectSchema = new mongoose.Schema({
  title:          { type: String, required: true, minlength: 3, maxlength: 200 },
  description:    { type: String, required: true, minlength: 20, maxlength: 2000 },
  studentIds:     [{ type: ObjectId, ref: 'Student' }],
  mentorId:       { type: ObjectId, ref: 'Teacher' },
  domain:         { type: String, required: true,
                    enum: ['AI/ML','Web Development','Mobile App Development',
                           'IoT','Cybersecurity','Blockchain','Data Science',
                           'Cloud Computing','DevOps','Game Development'] },
  batch:          { type: String, required: true },  // e.g. "2024-CSE"
  status:         { type: String, default: 'PROPOSED',
                    enum: ['PROPOSED','APPROVED','IN_PROGRESS',
                           'SUBMITTED','CHANGES_REQUESTED','GRADED'] },
  grade:          { type: Number, min: 0, max: 100 },
  submissionFile: { type: String },  // file path
  gradedAt:       { type: Date },
  gradedBy:       { type: ObjectId, ref: 'User' }
}, { timestamps: true });
```

10.1 Field-Level Database Specification

This section provides a field-level specification for the three most queried MongoDB collections, documenting each field's data type, constraints, default value, and index status.

User Collection — Field Specification

Field Name	BSON Type	Required	Unique	Default	Validation / Constraint
_id	ObjectId	Auto	Yes	Generated	Auto-generated by MongoDB
fullName	String	Yes	No	—	minlength=2, maxlength=100, trim
email	String	Yes	Yes	—	Regex email validation, lowercase
password	String	Yes	No	—	minlength=6, select:false (hidden)
role	String	Yes	No	—	Enum: ADMIN / TEACHER / STUDENT
isActive	Boolean	No	No	true	Soft-delete flag
profileImage	String	No	No	null	URL string, nullable
resetPasswordToken	String	No	No	undefined	select:false, used for password reset
resetPasswordExpires	Date	No	No	undefined	Expiry timestamp for reset token
createdAt	Date	Auto	No	Now	Mongoose timestamps: true
updatedAt	Date	Auto	No	Now	Mongoose timestamps: true

Student Collection — Field Specification

Field Name	BSON Type	Required	Unique	Default	Validation / Constraint
_id	ObjectId	Auto	Yes	Generated	Auto-generated
userId	ObjectId	Yes	Yes	—	ref: User, one-to-one
rollNo	String	Yes	Yes	—	uppercase, trim, indexed
course	String	Yes	No	—	trim, maxlength=200
semester	Number	Yes	No	—	min=1, max=10
section	String	Yes	No	—	uppercase, trim, maxlength=5
batchYear	String	No	No	—	Regex: ^\d{4}-[A-Z]{2,10}$
phone	String	No	No	—	Regex: phone number format
dateOfBirth	Date	No	No	—	Nullable date field
address	String	No	No	—	trim, maxlength=500
guardianName	String	No	No	—	trim, optional
guardianPhone	String	No	No	—	trim, optional
admissionDate	Date	No	No	Date.now	Defaults to creation date
isActive	Boolean	No	No	true	Soft-delete via deactivation
createdAt	Date	Auto	No	Now	Mongoose timestamps

Payment Collection — Field Specification

Field Name	BSON Type	Required	Unique	Default	Validation / Constraint
_id	ObjectId	Auto	Yes	Auto	Auto-generated
studentId	ObjectId	Yes	No	—	ref: Student
amount	Number	Yes	No	—	min=1 (must be positive)
type	String	Yes	No	—	Enum: tuition/hostel/library/lab/exam/transport/other
method	String	No	No	cash	Enum: cash/upi/netbanking/card/cheque/dd
transactionId	String	No	No	—	Optional reference ID
description	String	No	No	—	trim, maxlength=500
recordedBy	ObjectId	Yes	No	—	ref: User (admin who recorded)
receiptNo	String	No	Yes (sparse)	—	Auto-generated REC-YYYY-NNNNN
createdAt	Date	Auto	No	Now	Mongoose timestamps

Sparse unique index on receiptNo allows the field to be null/absent on older records without violating the unique constraint — only populated receipts are checked for uniqueness.

11. OUTPUT SCREENS OF THE PROJECT

The College Management System features a modern, responsive Single Page Application (SPA) interface built using React.js and styled with Tailwind CSS. The system supports both light and dark mode themes. The following screens describe the key user interface components across all three user roles.

1. Login Page

The login page provides secure role-based authentication. Users enter their email and password credentials. The system validates input, checks credentials against the database, and issues a JWT access token upon success. The interface includes a 'Forgot Password' link for password recovery and displays appropriate error messages for invalid credentials.

- Clean, minimal login form with email and password fields
- JWT token stored securely in localStorage upon authentication
- Role-based redirection: Admin → Admin Dashboard, Teacher → Teacher Dashboard, Student → Student Dashboard
- Password visibility toggle and validation feedback

2. Admin Dashboard

The Admin Dashboard provides a comprehensive overview of institutional statistics using real-time data from MongoDB aggregations. It features an AreaChart showing monthly revenue trends and a BarChart displaying student enrollment data.

- Total Students counter
- Total Teachers counter
- Total Revenue (monthly/cumulative)
- Attendance percentage overview
- Recent Admissions list
- Interactive revenue chart (last 6 months)

3. Student Management Module

The Student Management module allows administrators to register new students, view complete student profiles, update student information, and manage class assignments.

- Student registration form with full profile fields
- Search and filter students by course, semester, section
- View individual student profile with attendance percentage
- Edit student records and update semester/section assignments
- Export student list functionality

4. Teacher Management Module

The Teacher Management module handles faculty onboarding, assignment, and profile management.

- Add new teachers with department and designation assignment
- Assign subjects to teachers across courses and semesters
- View teacher profiles and subject allocations
- Update teacher designation and contact information

5. Attendance Management Module

The Attendance module provides a digital roll-call interface for teachers to mark daily attendance for their classes.

- Date-picker for selecting attendance date
- Class and subject selection dropdowns
- Bulk attendance marking (Present/Absent/Late) for all students
- Real-time attendance percentage calculation per student
- Attendance summary view with graphical representation

6. Fee Management Module

The Fee Management module handles financial transactions and maintains payment ledgers per student.

- Fee payment recording with amount, type, and date
- Student-wise fee history and outstanding balance display
- Payment status tracking (Pending/Completed/Failed)
- Admin revenue dashboard with monthly charts
- Fee receipt generation and export

7. Result Management Module

The Result Management module handles the entry and computation of student examination grades.

- Grade entry form for internal, external, and practical marks
- Automated result computation and pass/fail determination
- Student-facing result view with subject-wise breakdown
- Grade report generation per semester

8. Project Management Module

The Project Management module provides an end-to-end workflow for final year project tracking.

- Student project proposal form with domain selection
- Faculty mentor assignment and approval workflow
- Project status tracking (Proposed → Approved → In Progress → Submitted → Graded)
- PDF report upload for final project submission
- Faculty grading interface with grade and feedback entry
- Project export to CSV for department records

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11.2 User Interface Design Principles

The College Management System's user interface adheres to modern UX design principles to ensure usability, accessibility, and visual consistency across all modules.

Color Scheme & Typography

The system uses a dual-mode design with a carefully chosen color palette. The primary brand color is Indigo (#4F46E5) used for action buttons, active states, and accent elements. Supporting colors include green for success states, red for error/danger, amber for warnings, and purple for data highlights.

Color Token	Hex Value	Usage
Primary / Brand	#4F46E5	Buttons, active nav items, links, charts
Success / Green	#10B981	Attendance present, payment confirmed, pass status
Danger / Red	#EF4444	Absent, delete actions, error messages, fail status
Warning / Amber	#F59E0B	Pending states, late attendance, deadline alerts
Info / Cyan	#06B6D4	Informational badges, secondary highlights
Neutral Gray	#6B7280	Body text, subtitles, disabled states
Surface Light	#F9FAFB	Card backgrounds in light mode
Surface Dark	#1F2937	Card backgrounds in dark mode
Border Light	#E5E7EB	Card/table borders in light mode
Border Dark	#374151	Card/table borders in dark mode

Navigation Structure

The application uses a persistent left sidebar navigation with collapsible sections. The sidebar dynamically renders navigation items based on the authenticated user's role:

Role	Navigation Items Available
ADMIN	Dashboard, Students, Teachers, Courses, Classes, Attendance, Fee Management, Projects, Timetable, Reports, Settings

TEACHER	Dashboard, My Classes, Attendance, Timetable, Student Projects, Results, Profile
STUDENT	Dashboard, My Profile, Attendance, Fee History, My Project, Timetable, Results

Responsive Layout Breakpoints

The application uses Tailwind CSS v3 responsive prefixes to implement a fluid grid layout:

- **Mobile (default, < 640px):** Single-column stacked layout, collapsed sidebar (hamburger menu), full-width cards and tables with horizontal scroll.
- **Small (sm: 640px+):** Two-column grid for stat cards, sidebar overlay mode activated, condensed navigation labels.
- **Medium (md: 768px+):** Three-column stat grid, expanded sidebar with icons + labels, side-by-side form layout.
- **Large (lg: 1024px+):** Four-column stat grid, persistent sidebar, full data table display, chart panels side-by-side.
- **Extra Large (xl: 1280px+):** Maximum content width with centered layout, advanced dashboard widgets fully expanded.

Reusable Component Library

To maintain design consistency and reduce code duplication, the system implements a custom component library under src/components/ui/ with the following components:

Component	File	Props	Description
Button	ui/Button.jsx	variant, size, icon, disabled	Primary/secondary/danger styled button
Input	ui/Input.jsx	label, icon, error, type	Form input with label and validation state
Select	ui/Select.jsx	options, label, value, onChange	Styled dropdown with option mapping
Modal	ui/Modal.jsx	isOpen, onClose, title, size	Overlay modal with backdrop click-close
Badge	ui/Badge.jsx	color, size, text	Status pill badge with color variants
Card	ui/Card.jsx	title, subtitle, icon, actions	Content card with optional header
Table	ui/Table.jsx	columns, data, loading, onSort	Data table with sort and empty state
Spinner	ui/Spinner.jsx	size, color	Loading spinner for async operations
Alert	ui/Alert.jsx	type, message, onDismiss	Success/error/warning alert banner
PageHeader	PageHeader.jsx	title, subtitle, icon, actions	Consistent page header with breadcrumb

11B. SYSTEM IMPLEMENTATION

This chapter documents the complete implementation approach for the College Management System, covering the project setup, directory structure, development workflow, and deployment configuration.

11B.1 Project Setup & Initialization

The project was initialized using the following commands to scaffold both the frontend and backend:

Project Initialization Commands

```
# ■■ BACKEND SETUP ■■■
```

```
■   ■   ■■■ logo.png
■   ■■■ components/
■   ■   ■■■ Layout.jsx           # Main app shell with sidebar
■   ■   ■■■ Sidebar.jsx          # Role-aware navigation sidebar
■   ■   ■■■ Navbar.jsx           # Top navigation bar
■   ■   ■■■ PageHeader.jsx       # Reusable page title component
■   ■   ■■■ ProtectedRoute.jsx   # Auth guard wrapper
■   ■   ■■■ ui/
■   ■       ■■■ Button.jsx
■   ■       ■■■ Input.jsx
■   ■       ■■■ Select.jsx
■   ■       ■■■ Modal.jsx
■   ■       ■■■ Badge.jsx
■   ■       ■■■ Card.jsx
■   ■       ■■■ Table.jsx
■   ■       ■■■ Spinner.jsx
■   ■       ■■■ Alert.jsx
■   ■■■ context/
■   ■   ■■■ AuthContext.jsx      # Auth state provider
■   ■   ■■■ ThemeContext.jsx     # Dark/Light mode provider
■   ■■■ hooks/
■   ■   ■■■ useAuth.js           # Auth hook shorthand
■   ■   ■■■ useDebounce.js       # Search debounce utility
■   ■   ■■■ usePagination.js     # Pagination state hook
■   ■■■ pages/
■   ■   ■■■ Login.jsx            # Authentication page
■   ■   ■■■ AdminDashboard.jsx   # Admin overview
■   ■   ■■■ TeacherDashboard.jsx # Teacher overview
■   ■   ■■■ StudentDashboard.jsx # Student overview
■   ■   ■■■ Students.jsx         # Student CRUD
■   ■   ■■■ Teachers.jsx         # Teacher CRUD
■   ■   ■■■ Courses.jsx          # Course management
■   ■   ■■■ Classes.jsx          # Class & section management
■   ■   ■■■ Attendance.jsx       # Attendance marking
■   ■   ■■■ Fees.jsx             # Fee management
■   ■   ■■■ Projects.jsx         # Project lifecycle
■   ■   ■■■ Results.jsx          # Result management
■   ■   ■■■ Timetable.jsx        # Timetable viewer
■   ■   ■■■ Profile.jsx          # User profile page
■   ■■■ services/
■   ■   ■■■ api.js               # Axios instance + interceptors
■   ■■■ store/
■   ■   ■■■ authStore.js         # Zustand auth store
■   ■■■ utils/
■   ■   ■■■ formatters.js        # Date, currency formatters
■   ■   ■■■ validators.js        # Form validation helpers
■   ■■■ App.jsx                  # Root component + router
```

```
■   ■■■ main.jsx                # ReactDOM.createRoot entry
■   ■■■ index.css               # Tailwind base imports
■■■ .env                        # Vite environment variables
■■■ index.html                  # SPA root HTML
■■■ vite.config.js              # Vite + path alias config
■■■ tailwind.config.js          # Tailwind content paths
■■■ package.json
```

React Router Configuration (App.jsx)

App.jsx — React Router v6 Configuration

```
import { BrowserRouter, Routes, Route, Navigate } from 'react-router-dom';
import { ThemeProvider }   from '@/context/ThemeContext';
import { ProtectedRoute }  from '@/components/ProtectedRoute';
import { useAuthStore }    from '@/store/authStore';

// Pages
import { Login }            from '@/pages/Login';
import { AdminDashboard }   from '@/pages/AdminDashboard';
import { TeacherDashboard } from '@/pages/TeacherDashboard';
import { StudentDashboard } from '@/pages/StudentDashboard';
import { Students }         from '@/pages/Students';
import { Teachers }         from '@/pages/Teachers';
import { Attendance }       from '@/pages/Attendance';
import { Fees }             from '@/pages/Fees';
import { Projects }         from '@/pages/Projects';
import { Results }          from '@/pages/Results';
import { Timetable }        from '@/pages/Timetable';
import { Profile }          from '@/pages/Profile';

const RoleHome = () => {
  const { user } = useAuthStore();
  if (user?.role === 'ADMIN')    return <Navigate to="/admin"   replace />;
  if (user?.role === 'TEACHER')  return <Navigate to="/teacher" replace />;
  if (user?.role === 'STUDENT')  return <Navigate to="/student" replace />;
  return <Navigate to="/login" replace />;
};

export default function App() {
  return (
    <ThemeProvider>
      <BrowserRouter>
        <Routes>
          <Route path="/login" element={<Login />} />
          <Route path="/" element={
            <ProtectedRoute><RoleHome /></ProtectedRoute>
          } />

          {/* Admin Routes */}
          <Route path="/admin" element={
```

11B.2 MongoDB Index Strategy

To optimize query performance, the following indexes were created on MongoDB collections. Compound indexes were used on frequently filtered fields, and sparse indexes on optional fields.

Collection	Index	Type	Purpose
users	{ email: 1 }	Unique	Fast login lookup by email
students	{ userId: 1 }	Unique	One-to-one with users
students	{ rollNo: 1 }	Unique	Fast search by roll number
students	{ course: 1, semester: 1, section:1}	Compound	Filter students by class parameters
teachers	{ userId: 1 }	Unique	One-to-one with users
attendances	{ studentId: 1, date: -1 }	Compound	Student attendance history (latest first)
attendances	{ classId: 1, date: 1 }	Compound	Class attendance on a specific date
payments	{ studentId: 1, createdAt: -1 }	Compound	Student payment history
payments	{ createdAt: -1 }	Regular	Monthly revenue aggregation
projects	{ studentId: 1 }	Unique	One active project per student
projects	{ status: 1, domain: 1 }	Compound	Filter projects by status and domain
classes	{ course: 1, semester: 1, section:1}	Compound	Lookup class by course params

11B.3 Authentication & Session Flow

The College Management System implements a stateless JWT-based authentication system with access and refresh token rotation. The flow ensures secure session management without server-side session storage.

Complete Login & Token Refresh Flow

Step 1 — User submits credentials: User enters email + password on Login page and clicks Sign In.

Step 2 — Frontend API call: React sends POST /api/v1/auth/login with JSON body {email, password}.

Step 3 — Backend validates credentials: Express controller queries MongoDB for the user, compares the password against the bcrypt hash using bcrypt.compare(). Returns 401 if invalid.

Step 4 — Token generation: On success, the server calls generateTokens({id, role}) to create a 24-hour access token and a 7-day refresh token, both signed with separate secrets.

Step 5 — Token storage: Frontend receives {accessToken, refreshToken, user} and stores them in Zustand (persisted to localStorage via zustand/middleware persist).

Step 6 — Authenticated requests: Axios request interceptor attaches Authorization: Bearer header automatically to every subsequent API call.

Step 7 — Token expiry handling: When a 401 is returned by any endpoint, the Axios response interceptor calls POST /auth/refresh with the refreshToken. New tokens are issued and the original request is automatically retried with the new access token.

Step 8 — Logout: Frontend calls POST /auth/logout (to invalidate refresh token if stored server-side) then clears Zustand store — all subsequent requests will be unauthenticated.

11B.4 Error Handling Architecture

A centralized error handling architecture was implemented across both frontend and backend to ensure consistent error reporting and graceful failure handling.

Backend — AppError Class & Global Handler

utils/errors.js — Centralized Error Handling

```
// utils/errors.js
class AppError extends Error {
  constructor(message, statusCode = 500, isOperational = true) {
    super(message);
    this.statusCode   = statusCode;
    this.isOperational = isOperational;
    Error.captureStackTrace(this, this.constructor);
  }
}

// Global error handler (last middleware in Express)
const globalErrorHandler = (err, req, res, next) => {
  const statusCode = err.statusCode || 500;
  const message    = err.isOperational
    ? err.message
    : 'Internal server error. Please try again.';

  // Log stack trace in development
  if (process.env.NODE_ENV === 'development') {
    console.error('ERROR:', err.stack);
  }

  // Mongoose validation error
  if (err.name === 'ValidationError') {
    const messages = Object.values(err.errors)
      .map(e => e.message).join('. ');
    return res.status(400).json({
      success: false, message: `Validation error: ${messages}`
    });
  }

  // Mongoose duplicate key
  if (err.code === 11000) {
    const field = Object.keys(err.keyValue)[0];
    return res.status(409).json({
      success: false,
      message: `${field} already exists. Please use a different value.`
    });
```

```
    }
    res.status(statusCode).json({ success: false, message });
  };
  module.exports = { AppError, globalErrorHandler };
```

11B.5 Additional Backend Source Code

Below are additional controller and utility implementations covering the attendance and dashboard modules.

Attendance Controller (controllers/attendance.js)

controllers/attendance.js — Attendance Controller

```
'use strict';
const Attendance = require('../models/Attendance');
const Student    = require('../models/Student');
const { AppError } = require('../utils/errors');

// POST /api/v1/attendance/bulk
exports.markBulkAttendance = async (req, res, next) => {
  try {
    const { classId, date, records } = req.body;
    if (!classId || !date || !records?.length)
      return next(new AppError('classId, date, and records required', 400));

    const attendanceDate = new Date(date);
    attendanceDate.setHours(0, 0, 0, 0);

    // Delete existing records for same class+date (re-marking support)
    await Attendance.deleteMany({
      classId, date: attendanceDate
    });

    const docs = records.map(r => ({
      studentId: r.studentId,
      classId,
      date:      attendanceDate,
      status:    r.status || 'absent',
      markedBy:  req.user._id
    }));

    await Attendance.insertMany(docs);

    const summary = records.reduce((acc, r) => {
      acc[r.status] = (acc[r.status] || 0) + 1;
      return acc;
    }, {});

    res.status(201).json({
      success: true,
      message: `Attendance marked for ${records.length} students`,
      data: { date, classId, count: records.length, summary }
    });
  } catch (err) { next(err); }
};

// GET /api/v1/attendance/my-summary (Student)
exports.getMyAttendanceSummary = async (req, res, next) => {
  try {
```

12. CODING (MERN STACK) - COMPREHENSIVE

Complete source code implementation of the College Management System covering database schemas, backend middleware, frontend components, and API routing.

12.1 User Authentication & RBAC Schema Model

File: backend/src/models/User.js

```
javascript
"use strict";
var __importDefault = (this && this.__importDefault) || function (mod) {
    return (mod && mod.__esModule) ? mod : { "default": mod };
};
Object.defineProperty(exports, "__esModule", { value: true });
exports.User = void 0;
const mongoose_1 = __importDefault(require("mongoose"));
const userSchema = new mongoose_1.default.Schema({
    email: {
        type: String,
        unique: true,
        required: true,
        lowercase: true,
        trim: true,
    },
    password: {
        type: String,
        required: true,
    },
    fullName: {
        type: String,
        required: true,
    },
    phone: {
        type: String,
    },
    status: {
        type: String,
        enum: ['active', 'inactive', 'suspended'],
        default: 'active',
    },
    avatar: {
        type: String,
    },
    lastLogin: {
        type: Date,
    },
    roleAssignments: [
        {
            roleId: {
                type: mongoose_1.default.Schema.Types.ObjectId,
                ref: 'Role',
            },
            scopeId: {
                type: mongoose_1.default.Schema.Types.ObjectId,
                ref: 'Class',
            },
            assignedAt: {
                type: Date,
                default: Date.now,
            },
            assignedBy: {
                type: mongoose_1.default.Schema.Types.ObjectId,
                ref: 'User',
            },
        },
    ],
    teacherProfileId: {
        type: mongoose_1.default.Schema.Types.ObjectId,
        ref: 'Teacher',
    },
    studentProfileId: {
        type: mongoose_1.default.Schema.Types.ObjectId,
        ref: 'Student',
    },
}, { timestamps: true });
```

```
// Index for email
exports.User = mongoose_1.default.model('User', userSchema);
```

12.2 Student Demographics & Records Schema Model

File: backend/src/models/Student.js

```
javascript
"use strict";
var __importDefault = (this && this.__importDefault) || function (mod) {
    return (mod && mod.__esModule) ? mod : { "default": mod };
};
Object.defineProperty(exports, "__esModule", { value: true });
exports.Student = void 0;
const mongoose_1 = __importDefault(require("mongoose"));
const studentSchema = new mongoose_1.default.Schema({
    userId: {
        type: mongoose_1.default.Schema.Types.ObjectId,
        ref: 'User',
        required: true,
    },
    parentId: {
        type: mongoose_1.default.Schema.Types.ObjectId,
        ref: 'Parent',
    },
    rollNo: {
        type: String,
        unique: true,
        required: true,
    },
    enrollmentYear: {
        type: Number,
        required: true,
    },
    course: {
        type: String,
        required: true,
    },
    semester: {
        type: String,
        required: true,
    },
    section: {
        type: String,
        required: true,
    },
    dateOfBirth: {
        type: Date,
    },
    guardianInfo: {
        fatherName: String,
        motherName: String,
        fatherPhone: String,
        motherPhone: String,
        emergencyContact: String,
    },
    address: {
        street: String,
        city: String,
        state: String,
        pinCode: String,
        country: String,
    },
    status: {
        type: String,
        enum: ['active', 'inactive', 'graduated', 'suspended'],
        default: 'active',
    },
}, { timestamps: true });
// Indexes
studentSchema.index({ rollNo: 1 });
studentSchema.index({ course: 1 });
studentSchema.index({ userId: 1 });
exports.Student = mongoose_1.default.model('Student', studentSchema);
```

12.3 Faculty & Staff Information Schema Model

File: backend/src/models/Teacher.js

```
javascript
"use strict";
var __importDefault = (this && this.__importDefault) || function (mod) {
    return (mod && mod.__esModule) ? mod : { "default": mod };
};
Object.defineProperty(exports, "__esModule", { value: true });
exports.Teacher = void 0;
const mongoose_1 = __importDefault(require("mongoose"));
const teacherSchema = new mongoose_1.default.Schema({
    userId: {
        type: mongoose_1.default.Schema.Types.ObjectId,
        ref: 'User',
        required: true,
    },
    employeeId: {
        type: String,
        required: true,
        unique: true,
    },
    department: {
        type: String,
        required: true,
    },
    designation: {
        type: String,
        required: true,
    },
    qualification: {
        type: [String],
        required: true,
    },
    experience: {
        type: Number, // In years
        default: 0,
    },
    joiningDate: {
        type: Date,
        required: true,
    },
    subjects: [{
            type: String, // Can be refined to Subject model references later
        }],
    salary: {
        base: { type: Number, required: true },
        allowances: { type: Number, default: 0 },
        deductions: { type: Number, default: 0 },
    },
    status: {
        type: String,
        enum: ['active', 'on_leave', 'resigned', 'terminated'],
        default: 'active',
    },
    contactInfo: {
        phone: String,
        address: String,
    }
}, {
    timestamps: true,
});
exports.Teacher = mongoose_1.default.model('Teacher', teacherSchema);
```

12.4 Student Class Attendance Schema Model

File: backend/src/models/Attendance.js

```
javascript
'use strict';
const mongoose = require('mongoose');

const attendanceSchema = new mongoose.Schema({
    student: {
        type: mongoose.Schema.Types.ObjectId,
        ref: 'Student',
        required: true
    },
    class: {
        type: mongoose.Schema.Types.ObjectId,
        ref: 'Class',
        required: false
    },
    subject: {
        type: String,
        default: ''
    },
    date: {
        type: Date,
        required: true,
        default: Date.now
    },
    status: {
        type: String,
        enum: ['PRESENT', 'ABSENT', 'LATE', 'EXCUSED'],
        required: true,
        default: 'PRESENT'
    },
    markedBy: {
        type: mongoose.Schema.Types.ObjectId,
        ref: 'User',
        required: false
    },
    remarks: {
        type: String,
        default: ''
    }
}, {
    timestamps: true
});

// Compound index to prevent duplicate entries per student per date per subject
attendanceSchema.index({ student: 1, date: 1, subject: 1 }, { unique: false });

module.exports = mongoose.model('Attendance', attendanceSchema);
```

12.5 Academic Subject Entity Schema Model

File: backend/src/models/Subject.js

```
javascript
"use strict";
var __importDefault = (this && this.__importDefault) || function (mod) {
    return (mod && mod.__esModule) ? mod : { "default": mod };
};
Object.defineProperty(exports, "__esModule", { value: true });
exports.Subject = void 0;
const mongoose_1 = __importDefault(require("mongoose"));
const subjectSchema = new mongoose_1.default.Schema({
    name: {
        type: String,
        required: true,
    },
    code: {
        type: String,
        unique: true,
        required: true,
    },
    description: {
        type: String,
    },
    credits: {
        type: Number,
```

```
        },
        teachers: [
            {
                type: mongoose_1.default.Schema.Types.ObjectId,
                ref: 'User',
            },
        ],
    }, { timestamps: true });
    // Indexes
    subjectSchema.index({ code: 1 });
    exports.Subject = mongoose_1.default.model('Subject', subjectSchema);
```

12.6 System-wide Notification & Alerts Schema Model

File: backend/src/models/Notification.js

```
javascript
'use strict';
const mongoose = require('mongoose');

const notificationSchema = new mongoose.Schema({
    title: {
        type: String,
        required: true,
        trim: true,
    },
    message: {
        type: String,
        required: true,
    },
    type: {
        type: String,
        enum: ['INFO', 'WARNING', 'SUCCESS', 'ERROR', 'ALERT'],
        default: 'INFO',
    },
    targetRole: {
        type: String,
        enum: ['ADMIN', 'TEACHER', 'STUDENT', 'STAFF', 'ALL'],
        default: 'ALL',
    },
    isRead: {
        type: Boolean,
        default: false,
    },
    userId: {
        type: mongoose.Schema.Types.ObjectId,
        ref: 'User',
        default: null,
    },
    link: {
        type: String,
        default: null,
    },
}, {
    timestamps: true,
});

// Index for fast queries
notificationSchema.index({ isRead: 1, createdAt: -1 });
notificationSchema.index({ targetRole: 1, isRead: 1 });

const Notification = mongoose.model('Notification', notificationSchema);

module.exports = Notification;
```

12.7 Academic Class Timetable Schema Model

File: backend/src/models/Timetable.js

```
javascript
"use strict";
var __importDefault = (this && this.__importDefault) || function (mod) {
    return (mod && mod.__esModule) ? mod : { "default": mod };
};
Object.defineProperty(exports, "__esModule", { value: true });
exports.Timetable = void 0;
const mongoose_1 = __importDefault(require("mongoose"));
const timetableSchema = new mongoose_1.default.Schema({
    classId: {
        type: mongoose_1.default.Schema.Types.ObjectId,
        ref: 'Class',
        required: true,
    },
    academicYear: {
        type: String,
        required: true,
    },
    schedule: [
        {
            day: {
                type: String,
                enum: ['Monday', 'Tuesday', 'Wednesday', 'Thursday', 'Friday', 'Saturday'],
            },
            periods: [
                {
                    periodNo: Number,
                    startTime: String,
                    endTime: String,
                    subjectId: mongoose_1.default.Schema.Types.ObjectId,
                    teacherId: mongoose_1.default.Schema.Types.ObjectId,
                    room: String,
                },
            ],
        },
    ],
    status: {
        type: String,
        enum: ['draft', 'published'],
        default: 'draft',
    },
}, { timestamps: true });
// Indexes
timetableSchema.index({ classId: 1, academicYear: 1 });
exports.Timetable = mongoose_1.default.model('Timetable', timetableSchema);
```

12.8 Examination Scheduling Schema Model

File: backend/src/models/Exam.js

```
javascript
const mongoose = require('mongoose');

const examSchema = new mongoose.Schema({
    title: {
        type: String,
        required: true,
        trim: true
    },
    date: {
        type: Date,
        required: true
    },
    time: {
        type: String,
        required: true
    },
    course: {
        type: String,
        required: true
    },
    subject: {
        type: String,
        default: 'General'
    },
```

```
    type: {
        type: String,
        enum: ['Written', 'Practical', 'Online', 'Viva'],
        default: 'Written'
    },
    createdBy: {
        type: mongoose.Schema.Types.ObjectId,
        ref: 'User'
    }
}, {
    timestamps: true
});

module.exports = mongoose.model('Exam', examSchema);
```

12.9 JWT Authentication Verification Middleware

File: backend/src/middleware/auth.js

```
javascript
"use strict";
Object.defineProperty(exports, "__esModule", { value: true });
exports.roleMiddleware = exports.authMiddleware = void 0;
const jwt_1 = require("../utils/jwt");
const errors_1 = require("../utils/errors");
const authMiddleware = async (req, res, next) => {
    var _a;
    try {
        const token = (_a = req.headers.authorization) === null || _a === void 0 ? void 0 : _a.split(' ')[1];
        if (!token) {
            throw new errors_1.AppError('No authentication token provided', 401, 'NO_TOKEN');
        }
        const decoded = (0, jwt_1.verifyAccessToken)(token);
        if (!decoded) {
            throw new errors_1.AppError('Invalid or expired token', 401, 'INVALID_TOKEN');
        }
        req.user = decoded;
        next();
    }
    catch (error) {
        if (error instanceof errors_1.AppError) {
            res.status(error.statusCode).json({ message: error.message, errorCode: error.errorCode });
        }
        else {
            res.status(401).json({ message: 'Unauthorized' });
        }
    }
};
exports.authMiddleware = authMiddleware;
const roleMiddleware = (allowedRoles) => {
    return (req, res, next) => {
        if (!req.user) {
            return res.status(401).json({ message: 'Unauthorized' });
        }
        const hasRole = req.user.roles.some((role) => allowedRoles.includes(role));
        if (!hasRole) {
            return res.status(403).json({ message: 'Forbidden' });
        }
        next();
    };
};
exports.roleMiddleware = roleMiddleware;
```

12.10 Frontend React Application Root Bootstrap

File: frontend/src/main.jsx

```
jsx
import React from 'react';
import ReactDOM from 'react-dom/client';
import App from './App.jsx';
import './index.css';
ReactDOM.createRoot(document.getElementById('root')).render(<React.StrictMode>
  <App />
</React.StrictMode>);
```

12.11 Frontend React Application Main Router Matrix

File: frontend/src/App.jsx

```
jsx
import React, { useEffect } from 'react';
import { BrowserRouter as Router, useRoutes } from 'react-router-dom';
import { QueryClientProvider, QueryClient } from '@tanstack/react-query';
import { Toaster } from 'react-hot-toast';
import { useAuthStore } from '@/store/auth';
import { routes } from '@/config/routes';
import { ErrorBoundary } from '@/components/ErrorBoundary';

const queryClient = new QueryClient({
    defaultOptions: {
        queries: {
            retry: 1,
            refetchOnWindowFocus: false,
        }
    }
});

import { ThemeProvider } from '@/context/ThemeContext';
import { AnimatePresence, motion } from 'framer-motion';

const AppRoutes = () => {
    const routing = useRoutes(routes);
    return (
        <AnimatePresence mode="wait">
            <motion.div
                key={window.location.pathname}
                initial={{ opacity: 0, scale: 0.99, filter: 'blur(10px)' }}
                animate={{ opacity: 1, scale: 1, filter: 'blur(0px)' }}
                exit={{ opacity: 0, scale: 1.01, filter: 'blur(10px)' }}
                transition={{ duration: 0.4, ease: [0.22, 1, 0.36, 1] }}
            >
                {routing}
            </motion.div>
        </AnimatePresence>
    );
};

const Spinner = () => (
    <div className="flex flex-col items-center justify-center h-screen bg-slate-50 dark:bg-slate-950 gap-6">
        <div className="relative w-20 h-20">
            <div className="absolute inset-0 border-4 border-primary/10 rounded-[2rem] animate-pulse"></div>
            <div className="absolute inset-0 border-t-4 border-primary rounded-[2rem] animate-spin-slow"></div>
        </div>
        <div className="text-center">
            <p className="text-[10px] font-black tracking-[0.4em] uppercase text-slate-400 animate-pulse">Initializi
        </div>
    </div>
);

function App() {
    const { hydrate, isInitialized } = useAuthStore();
    useEffect(() => {
        hydrate();
    }, [hydrate]);

    if (!isInitialized) return <Spinner />;

    return (
        <QueryClientProvider client={queryClient}>
            <ThemeProvider>
                <Router>
```

```
                <ErrorBoundary>
                    <React.Suspense fallback={<Spinner />}>
                        <AppRoutes />
                    </React.Suspense>
                </ErrorBoundary>
            </Router>
            <Toaster
                position="bottom-right"
                toastOptio
```

12.12 Global Tailwind CSS Utilities Definitions

File: frontend/src/index.css

```
css
@import url('https://fonts.googleapis.com/css2?family=Inter:wght@300;400;500;600;700;800;900&family=Outfit:wght@300

@tailwind base;
@tailwind components;
@tailwind utilities;

@layer base {
  :root {
    --primary: 99 102 241;
    --secondary: 139 92 246;
    --success: 16 185 129;
    --warning: 245 158 11;
    --danger: 239 68 68;
    --info: 59 130 246;

    --background: 248 250 252;
    --foreground: 15 23 42;

    --card: 255 255 255;
    --card-foreground: 15 23 42;

    --border: 226 232 240;
    --input: 226 232 240;
    --ring: 99 102 241;

    /* Role Themes - Light */
    --admin-gradient: linear-gradient(135deg, #7c3aed 0%, #4f46e5 100%);
    --teacher-gradient: linear-gradient(135deg, #0ea5e9 0%, #0891b2 100%);
    --student-gradient: linear-gradient(135deg, #10b981 0%, #06b6d4 100%);
  }

  .dark {
    --background: 2 6 23;
    --foreground: 248 250 252;

    --card: 15 23 42;
    --card-foreground: 248 250 252;

    --border: 30 41 59;
    --input: 30 41 59;
    --ring: 129 140 248;
  }

  body {
    @apply bg-slate-50 text-slate-900 transition-colors duration-500 min-h-screen relative overflow-x-hidden;
    font-family: 'Inter', sans-serif;
    background-image: radial-gradient(circle at 50% -20%, #e2e8f0 0%, transparent 60%);
  }

  .dark body {
    @apply bg-[#020617] text-slate-50;
    background-image:
      radial-gradient(circle at 50% -20%, #1e1b4b 0%, transparent 65%),
      linear-gradient(to bottom, #020617, #020617);
  }

  /* Dot Grid Pattern */
  body::before {
    content: '';
    position: fixed;
    top: 0;
    left: 0;
    width: 100%;
    height: 100%;
    background-image: radial-gradient(#cbd5e1 1px, transparent 1px);
```

```
    background-size: 40px 40px;
    opacity: 0.15;
    pointer-events: none;
    z-index: -1;
  }

  .dark body::before {
    background-image: radial-gradient(#1e293b 1px, transparent 1px);
    opacity: 0.4;
  }
}

@layer components {
  .glass-card {
    @apply bg-white/70 dark:bg-slate-900/40 backdrop-blur-xl border border-white/40 dark:border-white/5 shadow-2xl;
  }

  .animate-shimmer {
    background: linear-gradient(90deg, transparent, rgba(255,255,255,0.2), transparent);
    background-size: 200% 100%;
    animation: shimmer 2s infinite;
  }

  .animate-spin-slow {
    animation: spin 3s linear infinite;
  }

  .role-admin { @apply from-violet-600 to-indigo-600; }
  .role-teacher { @apply from-sky-500 to-cyan-600; }
  .role-student
```

12.13 Public CMS Entry Landing Page View

File: frontend/src/pages/LandingPage.jsx

```
jsx
import React, { useEffect, useState, useRef } from 'react';
import { PublicHeader } from '@/components/PublicHeader';
import { useNavigate } from 'react-router-dom';
import {
    ArrowRight, ArrowUp, BookOpen, Users, Award, Building2, GraduationCap,
    Globe, ChevronRight, MapPin, Activity,
    Phone, Mail, CheckCircle, Star, TrendingUp, Zap, Shield
} from 'lucide-react';
import { motion, AnimatePresence } from 'framer-motion';



const slides = [
    {
        id: 0,
        tagline: "51st Annual Day Celebration 2024",
        title: "L.N. Mishra",
        subtitle: "Institute",
        highlight: "of Excellence",
        desc: "Bihar's premier management and technology institute, shaping future leaders since 1973 through indust
        bg: "/assets/images/lnmi/s1.jpeg",
        badge: "Est. 1973",
        overlayFrom: "from-slate-950/85",
    },
    {
        id: 1,
        tagline: "State-of-the-Art Auditorium",
        title: "World-Class",
        subtitle: "Learning",
        highlight: "Infrastructure",
        desc: "Our modern auditorium hosts convocations, national guest lectures and seminars with 1200+ seating cap
        bg: "/assets/images/lnmi/s2.jpeg",
        badge: "1200+ Capacity",
        overlayFrom: "from-indigo-950/80",
    },
    {
        id: 2,
        tagline: "Campus Illuminated — Festive Celebrations",
        title: "A Campus",
        subtitle: "That Comes",
        highlight: "Alive at Night",
        desc: "Every festival, every milestone — LNMI comes alive with light and celebration, fostering a vibrant co
        bg: "/assets/images/lnmi/lnmi-night1.webp",
        badge: "Vibrant Community",
        overlayFrom: "from-slate-950/75",
    },
    {
        id: 3,
        tagline: "L.N. Mishra Institute — Illuminated",
        title: "Where Tradition",
        subtitle: "Meets",
        highlight: "Modern Vision",
        desc: "The iconic LNMI Administrative Block, beautifully lit, symbolizing 50+ years of academic brilliance a
        bg: "/assets/images/lnmi/lnmi-night2.webp",
        badge: "50+ Years",
        overlayFrom: "from-slate-950/70",
    },
    {
        id: 4,
        tagline: "Advanced Computing Facility",
        title: "Technology-Driven",
        subtitle: "Academic",
        highlight: "Environment",
        desc: "Equipped with 300+ modern workstations, our computing labs power research and hands-on learning in cu
```

12.14 User Secure Login Interface View

File: frontend/src/pages/LoginPage.jsx

```
jsx
import React, { useState, useEffect } from 'react';
import { useNavigate } from 'react-router-dom';
import { useAuthStore } from '@/store/auth';
```

```
import { useTheme } from '@/context/ThemeContext';
import toast from 'react-hot-toast';
import { apiClient } from '@/services/api';
import { motion, AnimatePresence } from 'framer-motion';
import {
    Lock, Mail, Shield, Eye, EyeOff, GraduationCap,
    BookOpen, UserCircle, ArrowRight, Sparkles, Activity,
    ChevronLeft, ShieldCheck, Zap
} from 'lucide-react';

export const LoginPage = () => {
    const [step, setStep] = useState('ROLE_SELECT'); // ROLE_SELECT or LOGIN_FORM
    const [selectedRole, setSelectedRole] = useState('STUDENT');
    const [email, setEmail] = useState('');
    const [password, setPassword] = useState('');
    const [showPassword, setShowPassword] = useState(false);
    const [loading, setLoading] = useState(false);

    const navigate = useNavigate();
    const { login } = useAuthStore();
    const { isDarkMode } = useTheme();

    const roles = [
        { id: 'ADMIN', title: 'Admin', icon: Shield, color: 'from-purple-500 to-indigo-600', description: 'Administ
        { id: 'TEACHER', title: 'Teacher', icon: BookOpen, color: 'from-sky-500 to-cyan-600', description: 'Faculty
        { id: 'STUDENT', title: 'Student', icon: GraduationCap, color: 'from-emerald-500 to-cyan-500', description:
    ];

    const activeRoleData = roles.find(r => r.id === selectedRole);

    const handleLogin = async (e) => {
        e.preventDefault();
        setLoading(true);
        try {
            const response = await apiClient.post('/auth/login', {
                email,
                password,
                role: selectedRole
            });
            const { user, tokens } = response.data.data;
            const userRoles = user.roles || [];

            if (!userRoles.includes(selectedRole)) {
                toast.error(`Access denied. You do not have ${selectedRole} permissions.`);
                setLoading(false);
                return;
            }

            login(user, tokens.accessToken, tokens.refreshToken);
            toast.success(`Welcome back, ${user.fullName.split(' ')[0]}.`);
            navigate('/dashboard');
        } catch (error) {
            toast.error(error.response?.data?.message || 'Authentication Failed');
        } finally {
            setLoading(false);
        }
    };

    return (
```

12.15 User & Admin Protected Analytics Dashboard View

File: frontend/src/pages/Dashboard.jsx

```
jsx
import React, { useEffect, useState } from 'react';
import { Layout } from '@/components/Layout';
import {
    Users, GraduationCap, DollarSign, TrendingUp, CreditCard,
    Calendar as CalendarIcon, ArrowRight, CheckCircle, XCircle,
    Zap, Activity, Target, Landmark, LayoutDashboard,
    PieChart, Clock, ShieldCheck, UserPlus
} from 'lucide-react';
import {
    AreaChart, Area, XAxis, YAxis, CartesianGrid, Tooltip,
    ResponsiveContainer, BarChart, Bar, Cell
} from 'recharts';
import { apiClient } from '@/services/api';
import { useTheme } from '@/context/ThemeContext';
import { useAuthStore } from '@/store/auth';
```

```
import { motion, AnimatePresence } from 'framer-motion';
import { StudentDashboard } from '@/pages/StudentDashboard';
import { TeacherDashboard } from '@/pages/TeacherDashboard';
import { StatCard } from '@/components/dashboard/StatCard';
import { PageHeader } from '@/components/PageHeader';

const containerVariants = {
    hidden: { opacity: 0 },
    visible: { opacity: 1, transition: { staggerChildren: 0.1, delayChildren: 0.2 } }
};

const itemVariants = {
    hidden: { y: 20, opacity: 0 },
    visible: { y: 0, opacity: 1, transition: { type: 'spring', stiffness: 100 } }
};

import { useNavigate } from 'react-router-dom';

export const Dashboard = () => {
    const navigate = useNavigate();
    const { user } = useAuthStore();
    const { isDarkMode, roleTheme } = useTheme();

    const role = user?.roles?.[0]?.toUpperCase();

    if (role === 'STUDENT') return <StudentDashboard />;
    if (role === 'TEACHER') return <TeacherDashboard />;

    const [stats, setStats] = useState({
        totalStudents: 0,
        totalTeachers: 0,
        totalRevenue: 0,
        pendingFees: 0,
        attendance: { total: 0, present: 0, absent: 0, presentPct: 0 },
        recentAdmissions: [],
        chartData: []
    });
    const [loading, setLoading] = useState(true);

    useEffect(() => {
        fetchAdminStats();
    }, []);

    const fetchAdminStats = async () => {
        try {
            const response = await apiClient.get('/dashboard/admin');
            if (response.data.success) {
                setStats(response.data.data);
            }
        } catch (error) {
            console.error('Failed to fetch dashboard stats', error);
        } finally {
            setLoading(false);
        }
    };

    if (loading) {
        return (
            <Layout>
                <div className="flex flex-col justify-center items-center h
```

12.16 Student Ledger Management Interface View

File: frontend/src/pages/StudentsPage.jsx

```
jsx
import React, { useState, useEffect } from 'react';
import { Layout } from '@/components/Layout';
import { Button, DataTable, Input, Modal } from '@/components';
import { apiClient } from '@/services/api';
import toast from 'react-hot-toast';
import { Plus, GraduationCap, Search, Filter, User, Mail, ShieldCheck, ArrowRight, Trash2 } from 'lucide-react';
import { useNavigate } from 'react-router-dom';
import { PageHeader } from '@/components/PageHeader';
import { useTheme } from '@/context/ThemeContext';
import { useAuthStore } from '@/store/auth';
import { motion } from 'framer-motion';

export const StudentsPage = () => {
    const navigate = useNavigate();
    const { isDarkMode, roleTheme } = useTheme();
    const { user } = useAuthStore();
    const isAdmin = user?.roles?.includes('ADMIN');
    const [students, setStudents] = useState([]);
    const [loading, setLoading] = useState(false);
    const [page, setPage] = useState(1);
    const [total, setTotal] = useState(0);
    const [searchTerm, setSearchTerm] = useState('');

    useEffect(() => {
        fetchStudents();
    }, [page, searchTerm]);

    const fetchStudents = async () => {
        setLoading(true);
        try {
            const response = await apiClient.get(`/students?page=${page}&size=25&search=${searchTerm}`);
            const d = response.data.data;
            if (d && d.students) {
                setStudents(d.students);
                setTotal(d.total || d.students.length);
            } else if (Array.isArray(d)) {
                setStudents(d);
                setTotal(d.length);
            } else {
                setStudents([]);
                setTotal(0);
            }
        } catch (error) {
            toast.error(error.response?.data?.message || 'Failed to fetch students');
            console.error('Fetch Students Error:', error);
        } finally {
            setLoading(false);
        }
    };

    const handleDelete = async (id) => {
        if (!window.confirm('Are you sure you want to delete this student?')) return;
        try {
            await apiClient.delete(`/students/${id}`);
            toast.success('Student deleted successfully');
            fetchStudents();
        } catch (error) {
            toast.error(error.response?.data?.message || 'Failed to delete student');
        }
    };

    const columns = [
        {
            key: 'name',
            label: 'Student Name',
            render: (v, row) => (
                <div cla
```

12.17 Faculty Directory Interface View

File: frontend/src/pages/TeachersPage.jsx

```
jsx
import React, { useState } from 'react';
```

```
import { Layout } from '@/components/Layout';
import { useResource } from '@/hooks/useResource';
import { Plus, Search, Mail, Phone, BookOpen, ArrowLeft, UserPlus, Edit2, Trash2, Filter, GraduationCap, Award, Cale
import { useNavigate } from 'react-router-dom';
import { motion, AnimatePresence } from 'framer-motion';
import { Button, Input, PageHeader } from '@/components';
import { useTheme } from '@/context/ThemeContext';

export const TeachersPage = () => {
    const [searchTerm, setSearchTerm] = useState('');
    const navigate = useNavigate();
    const { isDarkMode, roleTheme } = useTheme();
    const { data: teachers, loading, remove } = useResource({ endpoint: '/teachers' });

    const handleDelete = async (id) => {
        if (!window.confirm('Terminate faculty appointment from digital registry?')) return;
        await remove(id);
    };

    const filteredTeachers = teachers.filter(teacher =>
        (teacher.userId?.fullName?.toLowerCase() || '').includes(searchTerm.toLowerCase()) ||
        (teacher.employeeId?.toLowerCase() || '').includes(searchTerm.toLowerCase()) ||
        (teacher.department?.toLowerCase() || '').includes(searchTerm.toLowerCase())
    );

    return (
        <Layout>
            <div className="max-w-[1400px] mx-auto px-4 pb-12">
                <PageHeader
                    title="Faculty Directorate"
                    subtitle={`${teachers.length} academic professionals registered`}
                    icon={GraduationCap}
                    backTo="/dashboard"
                    actions={
                        <Button
                            variant="primary"
                            icon={Plus}
                            onClick={() => navigate('/teachers/add')}
                        >
                            Commission Faculty
                        </Button>
                    }
                />

                <div className="space-y-8">
                    {/* Intelligence & Navigation Grid */}
                    <div className="flex flex-col md:flex-row gap-4 mb-8">
                        <div className="flex-1 relative">
                            <Input
                                type="search"
                                placeholder="Quantum search by faculty name, ID, or tenure..."
                                value={searchTerm}
                                onCh
```

12.18 Daily Attendance Registration Interface View

File: frontend/src/pages/AttendancePage.jsx

```
jsx
import React, { useState, useEffect, useCallback } from 'react';
import { Layout } from '@/components/Layout';
import { useAuthStore } from '@/store/auth';
import { apiClient } from '@/services/api';
import {
    CheckCircle, XCircle, Clock, Search, Filter,
    Download, Calendar, Users, Loader2, Save,
    History, ClipboardCheck, LayoutDashboard,
    TrendingUp, UserCheck, UserX, AlertCircle
} from 'lucide-react';
import toast from 'react-hot-toast';
import { DataTable, PageHeader, Button } from '@/components';
import { StatCard } from '@/components/dashboard/StatCard';
import { useTheme } from '@/context/ThemeContext';
import { motion, AnimatePresence } from 'framer-motion';

const MOCK_STUDENTS = [
    { _id: '65a1b2c3d4e5f60001000001', name: 'Aarav Sharma', rollNo: 'BCA001', class: 'BCA - Sem 1' },
    { _id: '65a1b2c3d4e5f60001000002', name: 'Priya Patel', rollNo: 'BCA002', class: 'BCA - Sem 1' },
    { _id: '65a1b2c3d4e5f60001000003', name: 'Rahul Verma', rollNo: 'BCA003', class: 'BCA - Sem 1' },
    { _id: '65a1b2c3d4e5f60001000004', name: 'Sneha Gupta', rollNo: 'BBA001', class: 'BBA - Sem 1' },
```

```
    { _id: '65a1b2c3d4e5f60001000005', name: 'Amit Joshi', rollNo: 'BBA002', class: 'BBA - Sem 1' },
    { _id: '65a1b2c3d4e5f60001000006', name: 'Kavya Singh', rollNo: 'MCA001', class: 'MCA - Sem 1' },
    { _id: '65a1b2c3d4e5f60001000007', name: 'Rohan Mehta', rollNo: 'MBA001', class: 'MBA - Sem 2' },
    { _id: '65a1b2c3d4e5f60001000008', name: 'Diya Kapoor', rollNo: 'BSC001', class: 'BSc IT - Sem 2' },
];

const getToday = () => new Date().toISOString().split('T')[0];

const getLast30Days = () => {
    const dates = [];
    for (let i = 0; i < 30; i++) {
        const d = new Date();
        d.setDate(d.getDate() - i);
        dates.push(d.toISOString().split('T')[0]);
    }
    return dates;
};

export const AttendancePage = () => {
    const { isDarkMode } = useTheme();
    const { user } = useAuthStore();
    const [selectedClass, setSelectedClass] = useState('All');
    const [selectedDate, setSelectedDate] = useState(getToday());
    const [searchTerm, setSearchTerm] = useState('');
    const [attendance, setAttendance] = useState({});
    const [saving, setSaving] = useState(false);
    const [loading, setLoading] = useState(true);
    const [submitted, setSubmitted] = useState(false);
    const [activeTab, setActiveTab] = useState('mark');
    const [allStudents, setAllStudents] = useState([]);
    const [classes, setClasses] = useState(['A
```

12.19 Financial Fee Monitoring Interface View

File: frontend/src/pages/FeesPage.jsx

```
jsx
import React, { useState, useEffect } from 'react';
import { Layout } from '@/components/Layout';
import { apiClient } from '@/services/api';
import { Plus, CreditCard, ChevronRight, X, IndianRupee, Layers, CalendarClock, Tag, ShieldCheck, TrendingUp, Wallet
import toast from 'react-hot-toast';
import { motion, AnimatePresence } from 'framer-motion';
import { PageHeader, Button } from '@/components';
import { StatCard } from '@/components/dashboard/StatCard';
import { useTheme } from '@/context/ThemeContext';
import { useAuthStore } from '@/store/auth';

const FREQUENCY_LABELS = {
    yearly: { label: 'Yearly', color: 'bg-indigo-500/10 text-indigo-500', icon: CalendarClock },
    semester: { label: 'Semester', color: 'bg-blue-500/10 text-blue-500', icon: Layers },
    monthly: { label: 'Monthly', color: 'bg-amber-500/10 text-amber-500', icon: TrendingUp },
    'one-time': { label: 'One-time', color: 'bg-emerald-500/10 text-emerald-500', icon: CreditCard },
};

const formatINR = (amount) =>
    new Intl.NumberFormat('en-IN', { maximumFractionDigits: 0 }).format(amount);

export const FeesPage = () => {
    const { isDarkMode } = useTheme();
    const { user } = useAuthStore();
    const isAdmin = user?.roles?.includes('ADMIN');

    const [feeTypes, setFeeTypes] = useState([]);
    const [loading, setLoading] = useState(true);
    const [showCreateModal, setShowCreateModal] = useState(false);

    // Assignment State
    const [showAssignModal, setShowAssignModal] = useState(false);
    const [selectedFeeForAssign, setSelectedFeeForAssign] = useState(null);
    const [classes, setClasses] = useState([]);
    const [selectedClassId, setSelectedClassId] = useState('');
    const [dueDate, setDueDate] = useState('');

    // Create Form State
    const [newFee, setNewFee] = useState({
        name: '',
        amount: '',
        frequency: 'yearly',
        description: '',
        isOptional: false,
    });

    useEffect(() => {
        fetchFeeTypes();
        if (isAdmin) fetchClasses();
    }, [isAdmin]);

    const fetchClasses = async () => {
        try {
            const response = await apiClient.get('/classes');
            if (response.data.success) setClasses(response.data.data);
        } catch {
        }
    };

    const fetchFeeTypes = async () => {
        try {
            const response = await apiClient.get('/finance/structure');
            if (response.data.success) setFeeTypes(response.data.data);
```

12.20 Interactive Drag-and-Drop Timetable Editor View

File: frontend/src/pages/TimetablePage.jsx

```
jsx
import React, { useState, useEffect } from 'react';
import { Layout } from '@/components/Layout';
import { Calendar, Plus, Clock, User, BookOpen, GraduationCap, X, CalendarDays, ShieldCheck, MapPin } from 'lucide-
import toast from 'react-hot-toast';
import { apiClient } from '@/services/api';
```

```
import { schedules } from '@/data/mockSchedules';
import { motion, AnimatePresence } from 'framer-motion';
import { PageHeader, Button } from '@/components';
import { useTheme } from '@/context/ThemeContext';
import { useAuthStore } from '@/store/auth';

export const TimetablePage = () => {
    const { isDarkMode } = useTheme();
    const { user } = useAuthStore();
    const isAdmin = user?.roles?.includes('ADMIN');
    const [selectedProgram, setSelectedProgram] = useState('BCA-1');
    const [selectedSemester, setSelectedSemester] = useState(1);
    const [showAddSlot, setShowAddSlot] = useState(false);
    const [editingSlot, setEditingSlot] = useState(null);
    const [teachers, setTeachers] = useState([]);

    useEffect(() => {
        const fetchTeachers = async () => {
            try {
                const response = await apiClient.get('/teachers');
                if (response.data.success && Array.isArray(response.data.data) && response.data.data.length > 0) {
                    setTeachers(response.data.data);
                }
            } catch (error) {
                setTeachers([]);
            }
        };
        fetchTeachers();
    }, []);

    const [newSlot, setNewSlot] = useState({
        day: 'Monday',
        time: '09:00 AM',
        subject: '',
        teacher: '',
        room: ''
    });

    const programs = [
        { id: 'BCA-1', name: 'BCA - Year 1' },
        { id: 'BCA-2', name: 'BCA - Year 2' },
        { id: 'BCA-3', name: 'BCA - Year 3' },
        { id: 'BBA-1', name: 'BBA - Year 1' },
        { id: 'BBA-2', name: 'BBA - Year 2' },
        { id: 'BBA-3', name: 'BBA - Year 3' },
        { id: 'MBA-1', name: 'MBA - Year 1' },
        { id: 'MBA-2', name: 'MBA - Year 2' },
    ];

    const days = ['Monday', 'Tuesday', 'Wednesday', 'Thursday', 'Friday', 'Saturday'];
    const timeSlots = ['09:00 AM', '10:00 AM', '11:00 AM', '12:00 PM', '02:00 PM', '03:00 PM'];
    const semesters = [1, 2];
    const [schedule, setSchedule] = useState([]);

    useEffect(() => {
        const key = `${selectedProgram}-sem${selectedSemester}`;
        setSchedule(schedules[key] || schedules[selectedProgram] || []);
    }, [selected
```

12.21 Examination Calendar Editor View

File: frontend/src/pages/ExamsPage.jsx

```
jsx
import React, { useState, useEffect } from 'react';
import { Layout } from '@/components/Layout';
import { useAuthStore } from '@/store/auth';
import { useTheme } from '@/context/ThemeContext';
import { Calendar, Clock, BookOpen, Plus, ShieldCheck, Hourglass, CheckCircle2, CalendarDays, X } from 'lucide-reac
import toast from 'react-hot-toast';
import { apiClient } from '@/services/api';
import { motion, AnimatePresence } from 'framer-motion';
import { PageHeader, Button } from '@/components';
import { StatCard } from '@/components/dashboard/StatCard';

export const ExamsPage = () => {
    const { user } = useAuthStore();
    const { isDarkMode } = useTheme();
    const [isCreateModalOpen, setIsCreateModalOpen] = useState(false);
    const [exams, setExams] = useState([]);
```

```
const [isLoading, setIsLoading] = useState(true);
const [formData, setFormData] = useState({
    title: '', date: '', time: '', course: 'BCA'
});

const fetchExams = async () => {
    try {
        setIsLoading(true);
        const response = await apiClient.get('/exams');
        if (response.data && response.data.success) {
            setExams(response.data.data);
        }
    } catch (error) {
        toast.error('Failed to load exams');
    } finally {
        setIsLoading(false);
    }
};

useEffect(() => {
    fetchExams();
}, []);

const canCreateExam = user?.roles?.includes('ADMIN') || user?.roles?.includes('TEACHER');

const handleCreateExam = async (e) => {
    e.preventDefault();
    try {
        // Ensure we send classId which is often required in backend
        const payload = {
            ...formData,
            classId: formData.course // Assuming course field in UI maps to classId in back
        };
        const response = await apiClient.post('/exams', payload);
        if (response.data && response.data.success) {
            toast.success('Exam scheduled successfully');
            setIsCreateModalOpen(false);
            fetchExams();
            setFormData({ title: '', date: '', time: '', course: 'BCA' });
        }
    } catch (error) {
        toast.error(error.response?.data?.message || 'Failed to schedule exam');
    }
};

const inputCls = `w-full h-12 px-4 rounded-xl border text-sm font-bold outline-none transition-all focus:ring-4
    ${isDarkMode ? 'bg-slate-
```

12.22 Persisted Navigation Sidebar Drawer Component

File: frontend/src/components/Sidebar.jsx

```
jsx
import React, { useState, useEffect } from 'react';
import { useNavigate, useLocation } from 'react-router-dom';
import { useAuthStore } from '@/store/auth';
import { useTheme } from '@/context/ThemeContext';
import { motion, AnimatePresence } from 'framer-motion';
import {
    ChevronDown, ChevronRight, Menu, Home, LogOut, Users,
    GraduationCap, Calculator, ClipboardCheck, FileText,
    Settings, Briefcase, LayoutDashboard, CreditCard,
    PieChart, UserCircle, CheckSquare, Sparkles, ChevronLeft
} from 'lucide-react';

export const Sidebar = () => {
    const [collapsed, setCollapsed] = useState(false);
    const [expandedMenus, setExpandedMenus] = useState(['FACULTY & STAFF']);
    const navigate = useNavigate();
    const location = useLocation();
    const { user, logout } = useAuthStore();
    const { isDarkMode, roleTheme } = useTheme();

    const menuItems = {
        ADMIN: [
            { label: 'Dashboard', path: '/dashboard', icon: LayoutDashboard },
            { label: 'Students', path: '/students', icon: GraduationCap },
            {
                label: 'Faculty & Staff',
                icon: UserCircle,
                children: [
                    { label: 'Faculty', path: '/teachers', icon: Users },
                    { label: 'Staff', path: '/staff', icon: Briefcase },
                ]
            },
            { label: 'Courses', path: '/classes', icon: ClipboardCheck },
            { label: 'Attendance', path: '/attendance', icon: CheckSquare },
            { label: 'Fees', path: '/fees', icon: CreditCard },
            { label: 'Exams', path: '/exams', icon: FileText },
            { label: 'Timetable', path: '/timetable', icon: Calculator },
            { label: 'Payroll', path: '/payroll', icon: PieChart },
            { label: 'Settings', path: '/settings', icon: Settings },
        ],
        STUDENT: [
            { label: 'My Dashboard', path: '/student-dashboard', icon: LayoutDashboard },
            { label: 'My Timetable', path: '/timetable', icon: Calculator },
            { label: 'Exams', path: '/exams', icon: FileText },
            { label: 'Grades', path: '/grades', icon: GraduationCap },
            { label: 'Settings', path: '/settings', icon: Settings },
        ],
        TEACHER: [
            { label: 'My Dashboard', path: '/teacher-dashboard', icon: LayoutDashboard },
            { label: 'Students', path: '/students', icon: GraduationCap },
            { label: 'Attendance', path: '/attendan
```

12.23 Authenticated User Toolbar Interface Component

File: frontend/src/components/Header.jsx

```
jsx
import React, { useState } from 'react';
import { useNavigate } from 'react-router-dom';
import { useAuthStore } from '@/store/auth';
import { useTheme } from '@/context/ThemeContext';
import { motion, AnimatePresence } from 'framer-motion';
import { Bell, Sun, Moon, LogOut, User, Settings, ChevronDown } from 'lucide-react';

export const Header = () => {
    const [isProfileOpen, setIsProfileOpen] = useState(false);
    const navigate = useNavigate();
    const { user, logout } = useAuthStore();
    const { isDarkMode, toggleDarkMode } = useTheme();

    const handleLogout = () => {
        logout();
        navigate('/login');
    };
```

```
    return (
        <header className="sticky top-0 z-30 w-full h-16 bg-white/80 dark:bg-slate-950/80 backdrop-blur-md border-b
            <div className="flex justify-between items-center h-full px-6">
                <div className="flex items-center gap-4">
                    {/* Placeholder for page title or search if needed later */}
                    <div className="hidden md:block">
                        <h2 className="text-sm font-bold text-slate-500 uppercase tracking-widest">Dashboard Overvie
                    </div>
                </div>

                <div className="flex items-center gap-4">
                    {/* Theme Toggle */}
                    <motion.button
                        whileTap={{ rotate: 180, scale: 0.8 }}
                        onClick={toggleDarkMode}
                        className="p-2.5 rounded-xl bg-slate-100 dark:bg-slate-800 text-slate-600 dark:text-slate-40
                    >
                        {isDarkMode ? (
                            <Sun size={20} className="text-amber-400" />
                        ) : (
                            <Moon size={20} className="text-indigo-600" />
                        )}
                        <span className="absolute top-12 left-1/2 -translate-x-1/2 px-2 py-1 bg-slate-800 text-white
                            {isDarkMode ? 'Light Mode' : 'Dark Mode'}
                        </span>
                    </motion.button>

                    {/* Notifications */}
                    <button className="p-2.5 rounded-xl bg-slate-100 dark:bg-slate-800
```

12.24 Public Informational Mega-Menu Navbar Component

File: frontend/src/components/PublicHeader.jsx

```jsx
import React, { useState, useEffect } from 'react';
import { useNavigate, useLocation } from 'react-router-dom';
import { Menu, X, ChevronDown, Rocket, GraduationCap, Building2, TrendingUp, Users, BookOpen } from 'lucide-react';
import { motion, AnimatePresence } from 'framer-motion';

export const PublicHeader = () => {
    const navigate = useNavigate();
    const location = useLocation();
    const [isMenuOpen, setIsMenuOpen] = useState(false);
    const [scrolled, setScrolled] = useState(false);
    const [activeDropdown, setActiveDropdown] = useState(null);

    useEffect(() => {
        const handleScroll = () => setScrolled(window.scrollY > 20);
        window.addEventListener('scroll', handleScroll);
        return () => window.removeEventListener('scroll', handleScroll);
    }, []);

    const navLinks = [
        {
            label: 'About',
            href: '#about',
            icon: Building2,
            subItems: [
                { title: 'History & Legacy', desc: 'Established in 1973', href: '#about-history' },
                { title: 'Infrastructure', desc: 'State-of-the-art facilities', href: '#about-infra' },
                { title: 'Leadership', desc: 'Visionary guidance', href: '#about-leadership' },
            ]
        },
        {
            label: 'Academics',
            href: '#academics',
            icon: GraduationCap,
            subItems: [
                { title: 'Undergraduate', desc: 'BBA & BCA Programs', href: '#ug-programs' },
                { title: 'Postgraduate', desc: 'MBA & MCA Programs', href: '#pg-programs' },
                { title: 'Integrated Courses', desc: 'BBA+MBA & BCA+MCA', href: '#integrated-programs' },
            ]
        },
        {
            label: 'Admissions',
            href: '#admissions',
            icon: BookOpen,
            subItems: [
                { title: 'Application Process', desc: 'Online registration (Jan-Jun)', href: '#apply' },
```

```
        { title: 'Eligibility Criteria', desc: 'UG & PG Requirements', href: '#eligibility' },
        { title: 'Entrance Exams', desc: 'MAT/CMAT/CAT & Internal Tests', href: '#exams' },
    ]
},
{
    label: 'Research',
    href: '#research',
    icon: TrendingUp,
    subItems: [
        { title: 'Socio-Economic Focus', desc: 'Legacy of societal research', href: '#research-focus' },
        { title: 'National Projects', desc: 'Government & in
```

12.25 Public Homepage Features Advertisement Component

File: frontend/src/components/AcademicsSection.jsx

```
jsx
import React, { useRef, useEffect } from 'react';
import gsap from 'gsap';
import { ScrollTrigger } from 'gsap/ScrollTrigger';
import { BookOpen, Award, Users, Globe } from 'lucide-react';
gsap.registerPlugin(ScrollTrigger);
const AcademicCard = ({ icon: Icon, title, description, delay }) => (<div className="academic-card bg-white p-8 rou
        <div className="w-14 h-14 bg-purple-100 rounded-full flex items-center justify-center mb-6 text-purple-600">
            <Icon size={28}/>
        </div>
        <h3 className="text-xl font-bold mb-3 text-gray-800">{title}</h3>
        <p className="text-gray-600 leading-relaxed">
            {description}
        </p>
    </div>);
export const AcademicsSection = () => {
    const sectionRef = useRef(null);
    useEffect(() => {
        const ctx = gsap.context(() => {
            gsap.from('.academic-card', {
                scrollTrigger: {
                    trigger: sectionRef.current,
                    start: 'top 80%',
                },
                y: 50,
                opacity: 0,
                duration: 0.8,
                stagger: 0.2,
                ease: "power3.out"
            });
        }, sectionRef);
        return () => ctx.revert();
    }, []);
    return (<section ref={sectionRef} className="py-24 bg-gray-50 relative z-20">
            <div className="container mx-auto px-6">
                <div className="text-center max-w-3xl mx-auto mb-16">
                    <span className="text-purple-600 font-semibold tracking-wider uppercase text-sm">World Class Edu
                    <h2 className="text-4xl font-bold mt-2 mb-4 text-gray-900">Academic Excellence</h2>
                    <p className="text-lg text-gray-600">
                        Our rigorous curriculum and innovative teaching methods prepare students for leadership in a
                    </p>
                </div>

                <div className="grid md:grid-cols-2 lg:grid-cols-4 gap-8">
                    <AcademicCard icon={BookOpen} title="Diverse Programs" description="Choose from over 50 undergra
                    <AcademicCard icon={Globe} title="Global Exposure" description="International partnerships and e
```

12.26 Re-usable Sortable Data Grid Component

File: frontend/src/components/DataTable.jsx

```
jsx
import React from 'react';
import { useTheme } from '@/context/ThemeContext';
import { Loader2, Database, SearchX } from 'lucide-react';
import { motion, AnimatePresence } from 'framer-motion';

export const DataTable = ({
    columns,
    data,
    loading,
    emptyIcon: EmptyIcon,
    emptyTitle = "No data found",
    emptyDescription = "There are no records to display.",
    onRowClick,
    tableContainerClassName = "",
    headerClassName = "",
    rowClassName = "",
}) => {
    const { isDarkMode } = useTheme();

    if (loading) {
        return (
            <div className="flex flex-col items-center justify-center py-24 gap-4">
                <div className="relative">
```

```
                    <div className="w-12 h-12 border-4 border-primary/20 rounded-full" />
                    <div className="absolute inset-0 w-12 h-12 border-4 border-primary border-t-transparent rounded-
                </div>
                <div className="flex flex-col items-center">
                    <span className="text-sm font-black uppercase tracking-[0.2em] text-primary">Synchronizing</span
                    <span className={`text-[10px] font-bold uppercase tracking-widest ${isDarkMode ? 'text-slate-600
                </div>
            </div>
        );
    }

    if (!data || data.length === 0) {
        return (
            <motion.div
                initial={{ opacity: 0, y: 20 }}
                animate={{ opacity: 1, y: 0 }}
                className={`px-6 py-20 text-center flex flex-col items-center justify-center gap-6 ${isDarkMode ? 't
            >
                <div className={`p-8 rounded-[2rem] ${isDarkMode ? 'bg-slate-900 border-slate-800 shadow-2xl shadow-
                    <div className="absolute inset-0 bg-primary/5 opacity-0 group-hover:opacity-100 transition-opaci
                    {EmptyIcon ? <EmptyIcon size={48} className="text-primary/40" /> : <SearchX size={48} className=
                </div>
                <div className="max-w-xs">
                    <h3 className={`font-black uppercase tracking-widest text-sm mb-2 ${isDarkMode ? 'text-slate-200
                    <p className="text-xs font-bold opacity-60 leading-relaxed uppercase tracking-tighter">{emptyDes
```

12.27 Generic Overlay Modal Window Component

File: frontend/src/components/Modal.jsx

```
jsx
import React from 'react';
import { X } from 'lucide-react';
import { motion, AnimatePresence } from 'framer-motion';
import { useThemeStore } from '@/store/theme';
export const Modal = ({ open, title, children, onClose, footer, size = 'md' }) => {
    const { isDarkMode } = useThemeStore();
    const sizeClasses = {
        sm: 'max-w-sm',
        md: 'max-w-md',
        lg: 'max-w-lg'
    };
    return (<AnimatePresence>
      {open && (<div className="fixed inset-0 z-50 flex items-center justify-center p-4">
          <motion.div initial={{ opacity: 0 }} animate={{ opacity: 1 }} exit={{ opacity: 0 }} onClick={onClose} clas
          <motion.div initial={{ opacity: 0, scale: 0.95 }} animate={{ opacity: 1, scale: 1 }} exit={{ opacity: 0, s
            <div className={`flex justify-between items-center p-6 border-b ${isDarkMode ? 'border-gray-700' : 'bord
              <h2 className="text-xl font-bold">{title}</h2>
              <button onClick={onClose} className={`p-1 rounded-lg transition-colors ${isDarkMode ? 'hover:bg-gray-7
                <X size={20}/>
              </button>
            </div>
            <div className="p-6">{children}</div>
            {footer && (<div className={`p-6 border-t flex gap-4 justify-end ${isDarkMode ? 'border-gray-700' : 'bor
                {footer}
              </div>)}
          </motion.div>
        </div>)}
    </AnimatePresence>);
};
```

12.28 Standardized UI Button Component

File: frontend/src/components/Button.jsx

```
jsx
import React from 'react';
import { motion, AnimatePresence } from 'framer-motion';
import { Loader2 } from 'lucide-react';

export const Button = ({
    variant = 'primary',
    size = 'md',
    isLoading = false,
    children,
    disabled,
    className,
    icon: Icon,
    ...props
}) => {
    const baseStyles = 'relative inline-flex items-center justify-center font-black uppercase tracking-widest transi

    const variantStyles = {
        primary: 'bg-primary text-white shadow-lg shadow-primary/20 hover:shadow-primary/40 rounded-2xl',
        secondary: 'bg-slate-100 dark:bg-slate-800 text-slate-600 dark:text-slate-300 hover:bg-slate-200 dark:hover
        danger: 'bg-rose-500 text-white shadow-lg shadow-rose-500/20 hover:shadow-rose-500/40 rounded-2xl',
        outline: 'border-2 border-slate-200 dark:border-slate-800 text-slate-600 dark:text-slate-400 hover:border-pr
        glass: 'glass-card border border-white/20 text-white hover:bg-white/20 rounded-2xl',
        ghost: 'bg-transparent text-slate-500 hover:bg-slate-100 dark:hover:bg-slate-800 rounded-xl',
    };

    const sizeStyles = {
        xs: 'px-3 py-1.5 text-[10px] gap-1.5',
        sm: 'px-4 py-2 text-[11px] gap-2',
        md: 'px-6 py-3 text-xs gap-2.5',
        lg: 'px-8 py-4 text-sm gap-3',
        icon: 'p-2.5 rounded-xl',
    };

    return (
        <motion.button
            whileHover={!disabled && !isLoading ? { y: -2, scale: 1.02 } : {}}
            whileTap={!disabled && !isLoading ? { scale: 0.98 } : {}}
            className={`${baseStyles} ${variantStyles[variant]} ${sizeStyles[size]} ${className}`}
            disabled={disabled || isLoading}
            {...props}
        >
            {/* Hover Shine Effect */}
            <div className="absolute inset-0 bg-gradient-to-r from-transparent via-white/10 to-transparent -translat

            <AnimatePresence mode="wait">
                {isLoading ? (
                    <motion.div
                        key="loader"
                        initial={{ opacity: 0, scale: 0.8 }}
                        animate={{ opacity: 1, scale: 1 }}
                        exit={{ opacity: 0, scale: 0.8 }}
                        className="flex items-center gap-2"
                    >
```

12. CODING (MERN STACK)

This chapter presents the key source code components of the College Management System, organized into Frontend (React.js SPA) and Backend (Node.js/Express API) sections. The code follows modern JavaScript (ES6+) conventions with clean separation of concerns.

12.1 Backend Node.js / Express API Architecture

Directory Structure: D:\collegemanagementMERN\College Management\backend\

Backend Folder Structure

Backend Project Structure

```
backend/
■■■ src/
■   ■■■ config/
■   ■   ■■■ index.js          # Environment config (port, DB, JWT, SMTP)
■   ■■■ constants/
■   ■   ■■■ project.constants.js  # Project domain, status, validation rules
■   ■■■ controllers/
■   ■   ■■■ auth.js            # Login, Register, Refresh, Reset Password
■   ■   ■■■ student.js         # Student CRUD operations
■   ■   ■■■ teacher.js         # Teacher CRUD operations
■   ■   ■■■ attendance.js      # Bulk attendance marking & summaries
■   ■   ■■■ dashboard.js       # Admin/Teacher/Student stats aggregation
■   ■   ■■■ payment.js         # Fee payment recording & history
■   ■   ■■■ project.js         # Project lifecycle management
■   ■   ■■■ class.js           # Class & section management
■   ■   ■■■ timetable.js       # Timetable scheduling
■   ■■■ models/
■   ■   ■■■ User.js / Student.js / Teacher.js
■   ■   ■■■ Attendance.js / Payment.js / Project.js
■   ■   ■■■ Class.js / Subject.js / Course.js
■   ■■■ routes/
■   ■   ■■■ *.js               # Express route definitions
■   ■■■ middleware/
■   ■   ■■■ auth.js            # JWT verification middleware
■   ■■■ utils/
■       ■■■ jwt.js             # Token generation & verification
■       ■■■ password.js        # bcrypt hash & compare
■       ■■■ errors.js          # AppError class
■■■ index.js                   # Express app entry point
```

Server Entry Point (index.js)

index.js — Server Entry Point

```
"use strict";
const express   = require('express');
const cors      = require('cors');
const mongoose  = require('mongoose');
const { config } = require('./config');

const app = express();
app.use(cors({ origin: config.frontendUrl, credentials: true }));
app.use(express.json({ limit: '50mb' }));

// Routes
app.use('/api/v1/auth',       require('./routes/auth'));
app.use('/api/v1/students',   require('./routes/student'));
app.use('/api/v1/teachers',   require('./routes/teacher'));
app.use('/api/v1/attendance', require('./routes/attendance'));
app.use('/api/v1/payments',   require('./routes/payment'));
app.use('/api/v1/projects',   require('./routes/project'));
app.use('/api/v1/classes',    require('./routes/class'));
app.use('/api/v1/dashboard',  require('./routes/dashboard'));

// MongoDB Connection
mongoose.connect(config.mongodb.uri)
  .then(() => {
    console.log('MongoDB connected successfully');
    app.listen(config.port, () =>
      console.log(`Server running on port ${config.port}`)
    );
  })
  .catch(err => { console.error('DB Connection Error:', err); process.exit(1); });
```

JWT Utility (utils/jwt.js)

JWT Token Utility

```
const jwt = require('jsonwebtoken');
const { config } = require('../config');

const generateTokens = (payload) => {
  const accessToken = jwt.sign(payload, config.jwt.secret,
    { expiresIn: config.jwt.expiresIn });                    // 24h
  const refreshToken = jwt.sign(payload, config.jwt.refreshSecret,
    { expiresIn: config.jwt.refreshExpiresIn });             // 7d
  return { accessToken, refreshToken };
};

const verifyToken = (token) => {
  try { return jwt.verify(token, config.jwt.secret); }
  catch { return null; }
};

const verifyRefreshToken = (token) => {
  try { return jwt.verify(token, config.jwt.refreshSecret); }
```

```
  catch { return null; }
};

module.exports = { generateTokens, verifyToken, verifyRefreshToken };
```

Authentication Controller (controllers/auth.js)

Authentication Controller — Login & Token Refresh

```
const { User }    = require('../models/User');
const { Role }    = require('../models/Role');
const { hashPassword, comparePassword } = require('../utils/password');
const { generateTokens } = require('../utils/jwt');
const { AppError }       = require('../utils/errors');

// POST /api/v1/auth/login
const authLogin = async (req, res) => {
  try {
    const { email, password } = req.body;
    const user = await User.findOne({ email })
      .populate({ path: 'roleAssignments.roleId', model: 'Role' });
    if (!user)
      return res.status(401).json({ message: 'Invalid credentials' });

    const isMatch = await comparePassword(password, user.password);
    if (!isMatch)
      return res.status(401).json({ message: 'Invalid credentials' });

    const roles = user.roleAssignments.map(ra => ra.roleId.name);
    const validRoles = ['ADMIN','TEACHER','STUDENT'];
    if (!roles.some(r => validRoles.includes(r)))
      return res.status(403).json({ message: 'No valid role assigned' });

    const tokens = generateTokens({
      userId: user._id.toString(), email: user.email, roles
    });
    user.lastLogin = new Date();
    await user.save();

    res.json({ message: 'Login successful', data: {
      user: { id: user._id, email: user.email,
              fullName: user.fullName, roles },
      tokens
    }});
  } catch (error) {
    res.status(500).json({ message: 'Internal server error' });
  }
};

// POST /api/v1/auth/refresh
const authRefresh = async (req, res) => {
  try {
    const { refreshToken } = req.body;
    const decoded = verifyRefreshToken(refreshToken);
```

```
    if (!decoded) throw new AppError('Invalid refresh token', 401);
    const user = await User.findById(decoded.userId)
                          .populate('roleAssignments.roleId');
    const roles = user.roleAssignments.map(ra => ra.roleId.name);
    const tokens = generateTokens({
      userId: user._id.toString(), email: user.email, roles
    });
    res.json({ message: 'Token refreshed', data: { tokens } });
  } catch (error) {
    res.status(error.statusCode || 500).json({ message: error.message });
  }
};

module.exports = { authLogin, authRefresh };
```

Attendance Controller (controllers/attendance.js)

Attendance Controller

```
const Attendance = require('../models/Attendance');

// POST /api/v1/attendance/bulk
const markAttendanceBulk = async (req, res) => {
  try {
    const { date, records, subject = '', classId } = req.body;
    if (!records || !Array.isArray(records) || records.length === 0)
      return res.status(400).json({ message: 'records array is required' });

    const attendanceDate = date ? new Date(date) : new Date();
    const markedById    = req.user?.userId;

    // Remove existing attendance for same date/subject to prevent duplicates
    const start = new Date(attendanceDate); start.setHours(0,0,0,0);
    const end   = new Date(attendanceDate); end.setHours(23,59,59,999);
    const studentIds = records.map(r => r.studentId);
    await Attendance.deleteMany({
      student: { $in: studentIds }, date: { $gte: start, $lte: end }, subject
    });

    const docs = records.map(r => ({
      student:  r.studentId,
      class:    classId || null,
      subject,
      date:     attendanceDate,
      status:   r.status || 'PRESENT',
      markedBy: markedById,
      remarks:  r.remarks || ''
    }));
    const saved = await Attendance.insertMany(docs);
    res.status(201).json({ message: 'Attendance marked', count: saved.length });
  } catch (err) {
    res.status(500).json({ message: 'Failed to mark attendance', error: err.message });
```

```
  }
};

// GET /api/v1/attendance/summary/:studentId
const getStudentAttendanceSummary = async (req, res) => {
  try {
    const { studentId } = req.params;
    const [total, present, absent, late] = await Promise.all([
      Attendance.countDocuments({ student: studentId }),
      Attendance.countDocuments({ student: studentId, status: 'PRESENT' }),
      Attendance.countDocuments({ student: studentId, status: 'ABSENT' }),
      Attendance.countDocuments({ student: studentId, status: 'LATE' }),
    ]);
    res.json({
      studentId, total, present, absent, late,
      percentage: total > 0 ? ((present / total) * 100).toFixed(1) : 0
    });
  } catch (err) {
    res.status(500).json({ message: 'Failed to get summary' });
  }
};

module.exports = { markAttendanceBulk, getStudentAttendanceSummary };
```

Dashboard Controller (controllers/dashboard.js)

Dashboard Controller — Admin Stats

```
const Student   = require('../models/Student');
const Teacher   = require('../models/Teacher');
const Payment   = require('../models/Payment');
const Attendance = require('../models/Attendance');

// GET /api/v1/dashboard/admin
const getAdminDashboardStats = async (req, res) => {
  try {
    const [totalStudents, totalTeachers] = await Promise.all([
      Student.countDocuments(),
      Teacher.countDocuments(),
    ]);

    // Revenue chart: last 6 months aggregation
    const sixMonthsAgo = new Date();
    sixMonthsAgo.setMonth(sixMonthsAgo.getMonth() - 5);
    sixMonthsAgo.setDate(1);

    const chartDataRaw = await Payment.aggregate([
      { $match: { paymentDate: { $gte: sixMonthsAgo }, status: 'COMPLETED' }},
      { $group: { _id: { month: { $month: '$paymentDate' },
                         year:  { $year:  '$paymentDate' }},
                  revenue:  { $sum: '$amount' },
                  students: { $addToSet: '$studentId' }}},
```

```
      { $sort: { '_id.year': 1, '_id.month': 1 }}
    ]);

    const months = ['Jan','Feb','Mar','Apr','May','Jun',
                    'Jul','Aug','Sep','Oct','Nov','Dec'];
    const chartData = chartDataRaw.map(d => ({
      month:    months[d._id.month - 1],
      revenue:  d.revenue,
      students: d.students.length
    }));
    const totalRevenue = chartDataRaw.reduce((s, d) => s + d.revenue, 0);

    // Attendance overview (today)
    const today = new Date();
    const startOfDay = new Date(today); startOfDay.setHours(0,0,0,0);
    const endOfDay   = new Date(today); endOfDay.setHours(23,59,59,999);
    const [totalAtt, presentAtt] = await Promise.all([
      Attendance.countDocuments({ date: { $gte: startOfDay, $lte: endOfDay }}),
      Attendance.countDocuments({ date: { $gte: startOfDay, $lte: endOfDay },
                                  status: 'PRESENT' }),
    ]);

    res.json({ success: true, data: {
      totalStudents, totalTeachers, totalRevenue, chartData,
      attendance: { total: totalAtt, present: presentAtt,
        presentPct: totalAtt > 0 ? ((presentAtt/totalAtt)*100).toFixed(1) : 0 }
    }});
  } catch (err) {
    res.status(500).json({ success: false, message: 'Dashboard error' });
  }
};

module.exports = { getAdminDashboardStats };
```

12.2 Frontend React.js SPA Architecture

Directory Structure: D:\collegemanagementMERN\College Management\frontend\

Frontend Folder Structure

Frontend Project Structure

```
frontend/
■■■ src/
■   ■■■ components/
■   ■   ■■■ Layout.jsx         # Main layout with navigation sidebar
■   ■   ■■■ PageHeader.jsx     # Reusable page header component
■   ■   ■■■ ui/                # Reusable UI: Button, Input, Card, Badge
■   ■   ■■■ dashboard/
■   ■       ■■■ StatCard.jsx   # Dashboard statistic card component
■   ■■■ pages/
■   ■   ■■■ Dashboard.jsx      # Role-based dashboard routing
■   ■   ■■■ AdminDashboard.jsx # Admin stats, charts, recent admissions
■   ■   ■■■ TeacherDashboard.jsx # Teacher's class & attendance overview
■   ■   ■■■ StudentDashboard.jsx # Student's personal stats dashboard
■   ■   ■■■ Students.jsx       # Student CRUD management
■   ■   ■■■ Teachers.jsx       # Teacher CRUD management
■   ■   ■■■ Attendance.jsx     # Bulk attendance marking interface
■   ■   ■■■ Fees.jsx           # Fee payment management
■   ■   ■■■ Results.jsx        # Grade management and result viewing
■   ■   ■■■ Projects.jsx       # Project lifecycle management
■   ■   ■■■ Timetable.jsx      # Timetable scheduling interface
■   ■■■ services/
■   ■   ■■■ api.js             # Axios instance with JWT interceptors
■   ■   ■■■ API_BASE_URL.js    # Environment-aware base URL config
■   ■   ■■■ project.service.js # Project API service functions
■   ■■■ store/
■   ■   ■■■ auth.js            # Zustand auth state (user, token, login/logout)
■   ■   ■■■ theme.js           # Zustand theme store (dark/light mode)
■   ■■■ context/
■   ■   ■■■ ThemeContext.jsx   # Theme context provider
■   ■■■ App.jsx                # Root component, React Router configuration
■■■ index.html
■■■ vite.config.js             # Vite bundler configuration
```

API Client with JWT Interceptors (services/api.js)

API Client — Axios with JWT Auto-Refresh

```
import axios from 'axios';
import { API_BASE_URL } from './API_BASE_URL';

export const apiClient = axios.create({
  baseURL: API_BASE_URL,
```

```
  headers: { 'Content-Type': 'application/json' },
});

// Attach JWT token to every request
apiClient.interceptors.request.use((config) => {
  const token = localStorage.getItem('accessToken');
  if (token) config.headers.Authorization = `Bearer ${token}`;
  return config;
});

// Handle 401 → auto refresh token
apiClient.interceptors.response.use(
  (response) => response,
  async (error) => {
    if (error.response?.status === 401) {
      const refreshToken = localStorage.getItem('refreshToken');
      if (refreshToken) {
        try {
          const res = await apiClient.post('/auth/refresh', { refreshToken });
          localStorage.setItem('accessToken',
            res.data.data.tokens.accessToken);
          localStorage.setItem('refreshToken',
            res.data.data.tokens.refreshToken);
          return apiClient(error.config);  // Retry original request
        } catch {
          localStorage.removeItem('accessToken');
          localStorage.removeItem('refreshToken');
          window.location.href = '/';      // Force redirect to login
        }
      }
    }
    return Promise.reject(error);
  }
);
```

Auth Store using Zustand (store/auth.js)

Zustand Auth Store — Login/Logout/Hydrate

```
import { create } from 'zustand';

export const useAuthStore = create((set) => ({
  user:            null,
  accessToken:     null,
  refreshToken:    null,
  isAuthenticated: false,
  isInitialized:   false,

  login: (user, accessToken, refreshToken) => {
    localStorage.setItem('accessToken',  accessToken);
    localStorage.setItem('refreshToken', refreshToken);
```

```
      localStorage.setItem('user',         JSON.stringify(user));
      set({ user, accessToken, refreshToken, isAuthenticated: true });
    },
    logout: () => {
      localStorage.removeItem('accessToken');
      localStorage.removeItem('refreshToken');
      localStorage.removeItem('user');
      set({ user: null, accessToken: null,
            refreshToken: null, isAuthenticated: false });
    },
    hydrate: () => {
      const accessToken  = localStorage.getItem('accessToken');
      const refreshToken = localStorage.getItem('refreshToken');
      const user         = localStorage.getItem('user');
      if (accessToken && refreshToken && user) {
        set({ accessToken, refreshToken, user: JSON.parse(user),
              isAuthenticated: true, isInitialized: true });
      } else {
        set({ isInitialized: true });
      }
    },
  }));
```

Project Service (services/project.service.js)

Project Service — API Functions

```
  import axios from './api';
  const API_URL = '/projects';
  export const projectService = {
    // Create new project proposal (Student role)
    createProject: async (data) => {
      const response = await axios.post(API_URL, data);
      return response.data;
    },
    // Get current student's own project
    getMyProject: async () => {
      const response = await axios.get(`${API_URL}/my-project`);
      return response.data;
    },
    // Submit final project report PDF (Student)
    submitReport: async (id, fileData) => {
      const response = await axios.post(`${API_URL}/${id}/submit`, fileData);
      return response.data;
    },
    // Get all projects with filters (Faculty / Admin)
    getAllProjects: async (filters = {}) => {
```

```
    const response = await axios.get(API_URL, { params: filters });
    return response.data;
  },
  // Grade a project (Faculty)
  gradeProject: async (id, gradeData) => {
    const response = await axios.post(`${API_URL}/${id}/grade`, gradeData);
    return response.data;
  },
  // Export projects to CSV (Admin / Faculty)
  exportToCSV: async (filters = {}) => {
    const response = await axios.get(`${API_URL}/export/csv`,
      { params: filters, responseType: 'blob' });
    return response.data;
  },
};
```

Class Controller (controllers/class.js)

Class Controller — CRUD Operations

```
const { Class }  = require('../models/Class');
const { AppError } = require('../utils/errors');
// POST /api/v1/classes
const createClass = async (req, res) => {
  try {
    const { name, section, academicYear, classTeacher, subjects } = req.body;
    const existing = await Class.findOne({ name, section, academicYear });
    if (existing)
      throw new AppError('Class already exists for this year', 400);
    const newClass = await Class.create({
      name, section, academicYear,
      classTeacher: classTeacher || null,
      subjects: subjects || []
    });
    res.status(201).json({ success: true, data: newClass });
  } catch (error) {
    const status = error instanceof AppError ? error.statusCode : 500;
    res.status(status).json({ message: error.message });
  }
};
// GET /api/v1/classes
const getAllClasses = async (req, res) => {
  try {
    const classes = await Class.find()
      .populate('classTeacher', 'firstName lastName email')
      .sort({ academicYear: -1, name: 1, section: 1 });
    res.json({ success: true, data: classes });
```

```
  } catch (error) {
    res.status(500).json({ message: 'Failed to fetch classes' });
  }
};

// PUT /api/v1/classes/:id
const updateClass = async (req, res) => {
  try {
    const updated = await Class.findByIdAndUpdate(
      req.params.id, req.body, { new: true, runValidators: true }
    );
    if (!updated) throw new AppError('Class not found', 404);
    res.json({ success: true, data: updated });
  } catch (error) {
    const status = error instanceof AppError ? error.statusCode : 500;
    res.status(status).json({ message: error.message });
  }
};

// DELETE /api/v1/classes/:id
const deleteClass = async (req, res) => {
  try {
    const deleted = await Class.findByIdAndDelete(req.params.id);
    if (!deleted) throw new AppError('Class not found', 404);
    res.json({ success: true, message: 'Class deleted successfully' });
  } catch (error) {
    const status = error instanceof AppError ? error.statusCode : 500;
    res.status(status).json({ message: error.message });
  }
};

module.exports = { createClass, getAllClasses, updateClass, deleteClass };
```

12.3 API Routes

Route Definitions Summary

Method	Route	Controller	Auth	Description
POST	/api/v1/auth/login	authLogin	Public	User login, returns JWT tokens
POST	/api/v1/auth/register	authRegister	Public	Register Admin/Student account
POST	/api/v1/auth/refresh	authRefresh	Public	Refresh expired access token
POST	/api/v1/auth/reset-password	resetPassword	Public	Reset user password by email
GET	/api/v1/students	getAllStudents	Admin	Fetch paginated student list
POST	/api/v1/students	createStudent	Admin	Register new student
PUT	/api/v1/students/:id	updateStudent	Admin	Update student profile
DELETE	/api/v1/students/:id	deleteStudent	Admin	Remove student record
GET	/api/v1/teachers	getAllTeachers	Admin	Fetch all teacher records
POST	/api/v1/teachers	createTeacher	Admin	Register new faculty member
POST	/api/v1/attendance/bulk	markAttendanceBulk	Teacher	Bulk mark class attendance
GET	/api/v1/attendance/summary/:id	getStudentSummary	Any	Student attendance summary
GET	/api/v1/payments	getAllPayments	Admin	Fetch all payment records
POST	/api/v1/payments	recordPayment	Admin	Record fee payment
GET	/api/v1/projects	getAllProjects	Teacher/Admin	Fetch all projects with filters
POST	/api/v1/projects	createProject	Student	Create new project proposal
POST	/api/v1/projects/:id/submit	submitReport	Student	Submit project PDF report
POST	/api/v1/projects/:id/grade	gradeProject	Teacher	Assign grade to project
GET	/api/v1/dashboard/admin	getAdminStats	Admin	Admin dashboard statistics
GET	/api/v1/classes	getAllClasses	Admin	Fetch all class sections
POST	/api/v1/classes	createClass	Admin	Create new class section

12.4 Frontend React Components

This section presents the complete React.js frontend source code for key pages and components. The frontend is built with React 18 + Vite, Tailwind CSS v3, and Zustand for state management.

Student Management Page (pages/Students.jsx)

The Students page provides full CRUD operations for student records. It includes search/filter functionality, a modal form for adding and editing students, and paginated table display with action buttons.

Students.jsx — Student Management Page

```
import React, { useEffect, useState } from 'react';
import { Layout }     from '@/components/Layout';
import { PageHeader } from '@/components/PageHeader';
import { apiClient }  from '@/services/api';
import { useTheme }   from '@/context/ThemeContext';
import { motion, AnimatePresence } from 'framer-motion';
import { Users, UserPlus, Search,
         Edit, Trash2, Eye } from 'lucide-react';
import { Button } from '@/components/ui/Button';
import { Input  } from '@/components/ui/Input';

export const Students = () => {
  const { isDarkMode } = useTheme();
  const [students,    setStudents]    = useState([]);
  const [filtered,    setFiltered]    = useState([]);
  const [loading,     setLoading]     = useState(true);
  const [searchTerm,  setSearchTerm]  = useState('');
  const [showModal,   setShowModal]   = useState(false);
  const [editStudent, setEditStudent] = useState(null);
  const [formData,    setFormData]    = useState({
    fullName: '', email: '', rollNo: '',
    course: '', semester: '', section: '',
    phone: '', address: '', dateOfBirth: ''
  });

  useEffect(() => { fetchStudents(); }, []);

  useEffect(() => {
    const lower = searchTerm.toLowerCase();
    setFiltered(students.filter(s =>
      s.userId?.fullName?.toLowerCase().includes(lower) ||
      s.rollNo?.toLowerCase().includes(lower)           ||
      s.course?.toLowerCase().includes(lower)
    ));
  }, [searchTerm, students]);

  const fetchStudents = async () => {
    try {
      setLoading(true);
```

```
      const { data } = await apiClient.get('/students');
      setStudents(data.data || []);
      setFiltered(data.data || []);
    } catch (err) {
      console.error('Failed to fetch students:', err);
    } finally { setLoading(false); }
  };

  const handleSubmit = async (e) => {
    e.preventDefault();
    try {
      if (editStudent) {
        await apiClient.put(`/students/${editStudent._id}`, formData);
      } else {
        await apiClient.post('/students', formData);
      }
      setShowModal(false);
      setEditStudent(null);
      fetchStudents();
    } catch (err) {
      console.error('Submit error:', err);
    }
  };

  const handleDelete = async (id) => {
    if (!window.confirm('Delete this student?')) return;
    await apiClient.delete(`/students/${id}`);
    fetchStudents();
  };

  return (
    <Layout>
      <PageHeader title="Student Management"
        subtitle={`Total: ${filtered.length} students`}
        icon={<Users size={28} />}
      />
      <div className="flex gap-4 mb-6">
        <Input placeholder="Search by name, roll no, course..."
          value={searchTerm}
          onChange={e => setSearchTerm(e.target.value)}
          icon={<Search size={16} />}
          className="flex-1"
        />
        <Button onClick={() => { setEditStudent(null);
          setFormData({ fullName:'',email:'',rollNo:'',
            course:'',semester:'',section:'',
            phone:'',address:'',dateOfBirth:'' });
          setShowModal(true); }}
          icon={<UserPlus size={16} />}
```

```
          >Add Student</Button>
        </div>
        {loading ? <div className="text-center py-8">Loading...</div> : (
          <div className="overflow-x-auto rounded-lg border">
            <table className="min-w-full divide-y divide-gray-200">
              <thead className="bg-gray-50 dark:bg-gray-800">
                <tr>
                  {['Roll No','Name','Course','Semester',
                    'Section','Phone','Actions'].map(h => (
                    <th key={h} className="px-4 py-3 text-left text-xs
                      font-medium text-gray-500 uppercase">{h}</th>
                  ))}
                </tr>
              </thead>
              <tbody className="divide-y divide-gray-200">
                {filtered.map(s => (
                  <motion.tr key={s._id}
                    initial={{ opacity: 0 }} animate={{ opacity: 1 }}
                    className="hover:bg-gray-50 dark:hover:bg-gray-700"
                  >
                    <td className="px-4 py-3 text-sm">{s.rollNo}</td>
                    <td className="px-4 py-3 text-sm font-medium">
                      {s.userId?.fullName}</td>
                    <td className="px-4 py-3 text-sm">{s.course}</td>
                    <td className="px-4 py-3 text-sm">{s.semester}</td>
                    <td className="px-4 py-3 text-sm">{s.section}</td>
                    <td className="px-4 py-3 text-sm">{s.phone}</td>
                    <td className="px-4 py-3 text-sm">
                      <div className="flex gap-2">
                        <button onClick={() => {
                          setEditStudent(s);
                          setFormData({ fullName: s.userId?.fullName || '',
                            email: s.userId?.email || '', rollNo: s.rollNo || '',
                            course: s.course || '', semester: s.semester || '',
                            section: s.section || '', phone: s.phone || '',
                            address: s.address || '', dateOfBirth: s.dateOfBirth || ''
                          }); setShowModal(true);
                        }} className="p-1 text-blue-500 hover:text-blue-700">
                          <Edit size={16} /></button>
                        <button onClick={() => handleDelete(s._id)}
                          className="p-1 text-red-500 hover:text-red-700">
                          <Trash2 size={16} /></button>
                      </div>
                    </td>
                  </motion.tr>
                ))}
              </tbody>
```

```
            </table>
          </div>
        )}
      </Layout>
    );
  };
```

Attendance Management Page (pages/Attendance.jsx)

The Attendance page allows teachers to mark bulk attendance for their class. Students are fetched by class/section, displayed in a grid, and submitted in a single API call using POST /api/v1/attendance/bulk.

Attendance.jsx — Attendance Management Page

```
import React, { useEffect, useState } from 'react';
import { Layout }         from '@/components/Layout';
import { PageHeader }     from '@/components/PageHeader';
import { apiClient }      from '@/services/api';
import { CalendarCheck, Users, CheckCircle, XCircle } from 'lucide-react';
import { Button }         from '@/components/ui/Button';
import { Select }         from '@/components/ui/Select';

export const Attendance = () => {
  const [classes,      setClasses]      = useState([]);
  const [students,     setStudents]     = useState([]);
  const [selectedClass, setSelectedClass] = useState('');
  const [attendance,   setAttendance]   = useState({});
  const [date,         setDate]         = useState(
    new Date().toISOString().split('T')[0]
  );
  const [submitting,   setSubmitting]   = useState(false);
  const [submitted,    setSubmitted]    = useState(false);

  useEffect(() => { fetchClasses(); }, []);

  useEffect(() => {
    if (selectedClass) fetchStudents(selectedClass);
  }, [selectedClass]);

  const fetchClasses = async () => {
    const { data } = await apiClient.get('/classes/my-classes');
    setClasses(data.data || []);
  };

  const fetchStudents = async (classId) => {
    const { data } = await apiClient.get(
      `/students?classId=${classId}`);
    setStudents(data.data || []);
    const initial = {};
    (data.data || []).forEach(s => { initial[s._id] = 'present'; });
    setAttendance(initial);
  };
```

```
const toggleStatus = (id) => {
  setAttendance(prev => ({
    ...prev,
    [id]: prev[id] === 'present' ? 'absent' : 'present'
  }));
};

const handleSubmit = async () => {
  setSubmitting(true);
  try {
    const records = students.map(s => ({
      studentId: s._id,
      status: attendance[s._id] || 'absent'
    }));
    await apiClient.post('/attendance/bulk', {
      classId: selectedClass, date, records
    });
    setSubmitted(true);
    setTimeout(() => setSubmitted(false), 3000);
  } catch (err) {
    console.error('Attendance submit error:', err);
  } finally { setSubmitting(false); }
};

const presentCount = Object.values(attendance)
  .filter(v => v === 'present').length;

return (
  <Layout>
    <PageHeader title="Attendance Management"
      icon={<CalendarCheck size={28} />}
      subtitle="Mark daily attendance for your classes"
    />
    <div className="grid grid-cols-1 md:grid-cols-3 gap-4 mb-6">
      <Select label="Select Class"
        options={classes.map(c => ({
          value: c._id, label: `${c.course} - Sem ${c.semester} (${c.section})`
        }))}
        value={selectedClass}
        onChange={e => setSelectedClass(e.target.value)}
      />
      <input type="date" value={date}
        onChange={e => setDate(e.target.value)}
        className="border rounded-lg px-4 py-2"
        max={new Date().toISOString().split('T')[0]}
      />
      <div className="flex items-center gap-4 text-sm font-medium">
        <span className="text-green-600">Present: {presentCount}</span>
        <span className="text-red-500">
```

```
            Absent: {students.length - presentCount}</span>
          </div>
        </div>
        {students.length > 0 && (
          <>
            <div className="grid grid-cols-2 md:grid-cols-3 lg:grid-cols-4 gap-3 mb-6">
              {students.map(s => (
                <button key={s._id}
                  onClick={() => toggleStatus(s._id)}
                  className={`flex items-center gap-3 p-3 rounded-lg border-2
                    transition-all ${attendance[s._id] === 'present'
                      ? 'border-green-500 bg-green-50'
                      : 'border-red-400 bg-red-50'}`}
                >
                  {attendance[s._id] === 'present'
                    ? <CheckCircle size={20} className="text-green-600" />
                    : <XCircle    size={20} className="text-red-500" />
                  }
                  <div className="text-left">
                    <p className="text-xs font-semibold">
                      {s.userId?.fullName}</p>
                    <p className="text-xs text-gray-500">{s.rollNo}</p>
                  </div>
                </button>
              ))}
            </div>
            <Button onClick={handleSubmit} disabled={submitting} size="lg">
              {submitting ? 'Submitting...' : 'Submit Attendance'}
            </Button>
            {submitted && (
              <p className="mt-3 text-green-600 font-medium">
                Attendance saved successfully!</p>
            )}
          </>
        )}
      </Layout>
    );
  };
```

Fee Management Page (pages/Fees.jsx)

The Fee Management page lets admin record and track student fee payments. It integrates with Recharts to display a monthly revenue bar chart fetched from the admin dashboard endpoint.

Fees.jsx — Fee Management Page

```
import React, { useEffect, useState } from 'react';
import { Layout }     from '@/components/Layout';
```

```
import { PageHeader } from '@/components/PageHeader';
import { apiClient }  from '@/services/api';
import { DollarSign, TrendingUp, CreditCard, Search } from 'lucide-react';
import { BarChart, Bar, XAxis, YAxis, CartesianGrid,
         Tooltip, ResponsiveContainer } from 'recharts';
import { Button } from '@/components/ui/Button';
import { Input  } from '@/components/ui/Input';

export const Fees = () => {
  const [payments,    setPayments]    = useState([]);
  const [filtered,    setFiltered]    = useState([]);
  const [stats,       setStats]       = useState({});
  const [chartData,   setChartData]   = useState([]);
  const [search,      setSearch]      = useState('');
  const [showModal,   setShowModal]   = useState(false);
  const [formData,    setFormData]    = useState({
    studentId: '', amount: '', type: 'tuition',
    method: 'cash', description: ''
  });

  useEffect(() => { fetchData(); }, []);

  useEffect(() => {
    const lower = search.toLowerCase();
    setFiltered(payments.filter(p =>
      p.studentId?.userId?.fullName?.toLowerCase().includes(lower) ||
      p.studentId?.rollNo?.toLowerCase().includes(lower)
    ));
  }, [search, payments]);

  const fetchData = async () => {
    const [paymentsRes, dashRes] = await Promise.all([
      apiClient.get('/payments'),
      apiClient.get('/dashboard/admin')
    ]);
    setPayments(paymentsRes.data.data || []);
    setFiltered(paymentsRes.data.data || []);
    setStats(dashRes.data.data?.stats || {});
    setChartData(dashRes.data.data?.revenueChart || []);
  };

  const handleRecord = async (e) => {
    e.preventDefault();
    await apiClient.post('/payments', {
      ...formData,
      amount: Number(formData.amount)
    });
    setShowModal(false);
    fetchData();
  };
```

```
    const feeTypes = ['tuition','hostel','library',
                      'laboratory','exam','transport','other'];

    return (
      <Layout>
        <PageHeader title="Fee Management"
          icon={<DollarSign size={28} />}
          subtitle="Track student fee payments and revenue"
        />
        <div className="grid grid-cols-1 md:grid-cols-3 gap-4 mb-6">
          <div className="bg-white rounded-xl p-4 shadow border">
            <p className="text-sm text-gray-500">Total Revenue</p>
            <p className="text-2xl font-bold text-green-600">
              ■{(stats.totalRevenue || 0).toLocaleString()}</p>
          </div>
          <div className="bg-white rounded-xl p-4 shadow border">
            <p className="text-sm text-gray-500">This Month</p>
            <p className="text-2xl font-bold text-blue-600">
              ■{(stats.monthlyRevenue || 0).toLocaleString()}</p>
          </div>
          <div className="bg-white rounded-xl p-4 shadow border">
            <p className="text-sm text-gray-500">Transactions</p>
            <p className="text-2xl font-bold text-purple-600">
              {stats.totalPayments || 0}</p>
          </div>
        </div>
        <div className="bg-white rounded-xl p-4 shadow border mb-6">
          <h3 className="text-sm font-semibold mb-4">6-Month Revenue</h3>
          <ResponsiveContainer width="100%" height={200}>
            <BarChart data={chartData}>
              <CartesianGrid strokeDasharray="3 3" />
              <XAxis dataKey="month" tick={{ fontSize: 12 }} />
              <YAxis tick={{ fontSize: 12 }} />
              <Tooltip formatter={v => `■${v.toLocaleString()}`} />
              <Bar dataKey="revenue" fill="#4F46E5" radius={[4,4,0,0]} />
            </BarChart>
          </ResponsiveContainer>
        </div>
        <div className="flex gap-4 mb-4">
          <Input placeholder="Search by student name or roll no..."
            value={search}
            onChange={e => setSearch(e.target.value)}
            icon={<Search size={16} />} className="flex-1"
          />
          <Button onClick={() => setShowModal(true)}
            icon={<CreditCard size={16} />}>
            Record Payment</Button>
        </div>
```

```
        <div className="overflow-x-auto rounded-lg border">
          <table className="min-w-full divide-y divide-gray-200">
            <thead className="bg-gray-50">
              <tr>
                {['Student','Roll No','Amount','Type',
                  'Method','Date','Status'].map(h => (
                  <th key={h} className="px-4 py-3 text-left text-xs
                    font-medium text-gray-500 uppercase">{h}</th>
                ))}
              </tr>
            </thead>
            <tbody className="divide-y divide-gray-200">
              {filtered.map(p => (
                <tr key={p._id}
                  className="hover:bg-gray-50">
                  <td className="px-4 py-3 text-sm font-medium">
                    {p.studentId?.userId?.fullName}</td>
                  <td className="px-4 py-3 text-sm">
                    {p.studentId?.rollNo}</td>
                  <td className="px-4 py-3 text-sm font-semibold
                    text-green-600">■{p.amount?.toLocaleString()}</td>
                  <td className="px-4 py-3 text-sm capitalize">
                    {p.type}</td>
                  <td className="px-4 py-3 text-sm capitalize">
                    {p.method}</td>
                  <td className="px-4 py-3 text-sm">
                    {new Date(p.createdAt).toLocaleDateString()}</td>
                  <td className="px-4 py-3">
                    <span className="px-2 py-1 rounded-full text-xs
                      font-medium bg-green-100 text-green-700">Paid</span>
                  </td>
                </tr>
              ))}
            </tbody>
          </table>
        </div>
      </Layout>
    );
  };
```

Project Management Page (pages/Projects.jsx)

The Projects page handles the full project lifecycle. Faculty can view submitted projects, approve/reject proposals, and grade final submissions. Students can submit proposals and track project status.

Projects.jsx — Project Management Page

```
import React, { useEffect, useState, useRef } from 'react';
import { Layout }     from '@/components/Layout';
import { PageHeader } from '@/components/PageHeader';
import { apiClient }  from '@/services/api';
import { useAuth }    from '@/context/AuthContext';
import { motion, AnimatePresence } from 'framer-motion';
import { BookOpen, PlusCircle, CheckCircle,
         XCircle, Star, ChevronDown } from 'lucide-react';
import { Button } from '@/components/ui/Button';

const STATUS_COLORS = {
  PROPOSED:           'bg-yellow-100 text-yellow-800',
  APPROVED:           'bg-blue-100   text-blue-800',
  IN_PROGRESS:        'bg-indigo-100 text-indigo-800',
  SUBMITTED:          'bg-purple-100 text-purple-800',
  CHANGES_REQUESTED:  'bg-orange-100 text-orange-800',
  GRADED:             'bg-green-100  text-green-800',
};

const DOMAINS = [
  'AI/ML','Web Development','Mobile App Development','IoT',
  'Cybersecurity','Blockchain','Data Science','Cloud Computing',
  'DevOps','Game Development'
];

export const Projects = () => {
  const { user }    = useAuth();
  const isStudent   = user?.role === 'STUDENT';
  const isTeacher   = user?.role === 'TEACHER';
  const isAdmin     = user?.role === 'ADMIN';

  const [projects,    setProjects]    = useState([]);
  const [loading,     setLoading]     = useState(true);
  const [showPropose, setShowPropose] = useState(false);
  const [gradeModal,  setGradeModal]  = useState(null);
  const [gradeData,   setGradeData]   = useState({
    grade: '', feedback: ''
  });
  const [proposeData, setProposeData] = useState({
    title: '', description: '', domain: '',
    techStack: '', objectives: '', timeline: ''
  });

  useEffect(() => { fetchProjects(); }, []);

  const fetchProjects = async () => {
```

```
    try {
      setLoading(true);
      const endpoint = isStudent
        ? '/projects/my-project'
        : '/projects';
      const { data } = await apiClient.get(endpoint);
      const arr = isStudent
        ? (data.data ? [data.data] : [])
        : (data.data || []);
      setProjects(arr);
    } catch (err) {
      if (err.response?.status !== 404)
        console.error('Projects fetch error:', err);
    } finally { setLoading(false); }
  };

  const handlePropose = async (e) => {
    e.preventDefault();
    await apiClient.post('/projects', {
      ...proposeData,
      techStack: proposeData.techStack
        .split(',').map(t => t.trim()).filter(Boolean)
    });
    setShowPropose(false);
    fetchProjects();
  };

  const handleStatusChange = async (id, status) => {
    await apiClient.put(`/projects/${id}/status`, { status });
    fetchProjects();
  };

  const handleGrade = async (e) => {
    e.preventDefault();
    await apiClient.post(
      `/projects/${gradeModal._id}/grade`, gradeData
    );
    setGradeModal(null);
    fetchProjects();
  };

  return (
    <Layout>
      <PageHeader title="Project Management"
        icon={<BookOpen size={28} />}
        subtitle={isStudent
          ? 'Submit and track your academic project'
          : `All projects (${projects.length})`}
      />
      {isStudent && projects.length === 0 && (
```

```
          <Button onClick={() => setShowPropose(true)}
            icon={<PlusCircle size={16} />} className="mb-6">
            Propose New Project</Button>
        )}
        <div className="space-y-4">
          {projects.map(p => (
            <motion.div key={p._id}
              initial={{ opacity: 0, y: 10 }}
              animate={{ opacity: 1, y: 0 }}
              className="bg-white rounded-xl p-5 shadow border"
            >
              <div className="flex items-start justify-between">
                <div className="flex-1">
                  <h3 className="font-semibold text-gray-800 mb-1">
                    {p.title}</h3>
                  <p className="text-sm text-gray-500 mb-2">{p.domain}</p>
                  <p className="text-sm text-gray-600 mb-3">
                    {p.description}</p>
                  <div className="flex flex-wrap gap-2 mb-3">
                    {p.techStack?.map(t => (
                      <span key={t}
                        className="px-2 py-0.5 bg-gray-100 text-gray-600
                          text-xs rounded-full">{t}</span>
                    ))}
                  </div>
                  <div className="flex items-center gap-4 text-xs text-gray-500">
                    {!isStudent && (
                      <span>Student: {p.studentId?.userId?.fullName}</span>
                    )}
                    <span>
                      Submitted: {new Date(p.createdAt).toLocaleDateString()}
                    </span>
                    {p.grade && (
                      <span className="flex items-center gap-1 text-yellow-600">
                        <Star size={12} /> Grade: {p.grade}
                      </span>
                    )}
                  </div>
                </div>
                <div className="flex flex-col items-end gap-2 ml-4">
                  <span className={`px-3 py-1 rounded-full text-xs font-semibold
                    ${STATUS_COLORS[p.status] || 'bg-gray-100 text-gray-600'}`}>
                    {p.status?.replace('_',' ')}
                  </span>
                  {(isTeacher || isAdmin) &&
                    p.status === 'PROPOSED' && (
                    <div className="flex gap-2 mt-2">
```

```
                    <button onClick={() =>
                      handleStatusChange(p._id,'APPROVED')}
                      className="flex items-center gap-1 text-xs px-3 py-1
                        bg-green-100 text-green-700 rounded-full hover:bg-green-200">
                      <CheckCircle size={12}/> Approve</button>
                    <button onClick={() =>
                      handleStatusChange(p._id,'CHANGES_REQUESTED')}
                      className="flex items-center gap-1 text-xs px-3 py-1
                        bg-red-100 text-red-700 rounded-full hover:bg-red-200">
                      <XCircle size={12}/> Request Changes</button>
                  </div>
                )}
                {(isTeacher || isAdmin) &&
                  p.status === 'SUBMITTED' && (
                  <button onClick={() => setGradeModal(p)}
                    className="flex items-center gap-1 text-xs px-3 py-1
                      bg-yellow-100 text-yellow-700 rounded-full
                      hover:bg-yellow-200 mt-2">
                    <Star size={12}/> Grade</button>
                )}
              </div>
            </div>
          </motion.div>
        ))}
      </div>
    </Layout>
  );
};
```

12.5 Backend Middleware & Controllers

The backend layer consists of Express.js middleware for security and controllers that handle business logic. All controllers follow the same pattern: validate input, query MongoDB, return structured JSON responses.

JWT Authentication Middleware (middleware/auth.js)

Every protected API route passes through this middleware. It extracts the Bearer token from the Authorization header, verifies it, and attaches the decoded payload to req.user.

middleware/auth.js — JWT Authentication Middleware

```
'use strict';
const { verifyToken } = require('../utils/jwt');
const User = require('../models/User');

/**
 * Protect middleware — verifies JWT and attaches user to req
 * Usage: router.get('/protected', protect, handler)
 */
const protect = async (req, res, next) => {
  try {
    const authHeader = req.headers.authorization;
    if (!authHeader || !authHeader.startsWith('Bearer ')) {
      return res.status(401).json({
        success: false,
        message: 'Access denied. No token provided.'
      });
    }
    const token   = authHeader.split(' ')[1];
    const decoded = verifyToken(token);
    if (!decoded) {
      return res.status(401).json({
        success: false,
        message: 'Invalid or expired token.'
      });
    }
    const user = await User.findById(decoded.id)
      .select('-password -refreshToken');
    if (!user || !user.isActive) {
      return res.status(401).json({
        success: false,
        message: 'User not found or account deactivated.'
      });
    }
    req.user = user;
    next();
  } catch (error) {
    return res.status(500).json({
```

```
      success: false,
      message: 'Authentication error.',
      error: error.message
    });
  }
};

/**
 * Role-based authorization middleware
 * Usage: authorize('ADMIN', 'TEACHER')
 */
const authorize = (...roles) => {
  return (req, res, next) => {
    if (!roles.includes(req.user?.role)) {
      return res.status(403).json({
        success: false,
        message: `Role '${req.user?.role}' is not authorized.`
      });
    }
    next();
  };
};

module.exports = { protect, authorize };
```

Payment Controller (controllers/payment.js)

The payment controller handles fee recording, history retrieval, and revenue aggregation. It validates student existence before recording payments and supports filtering by type, method, and date range.

controllers/payment.js — Payment Controller

```
'use strict';
const Payment = require('../models/Payment');
const Student = require('../models/Student');
const { AppError } = require('../utils/errors');

// POST /api/v1/payments — Record new fee payment
exports.recordPayment = async (req, res, next) => {
  try {
    const { studentId, amount, type, method, description } = req.body;
    if (!studentId || !amount || !type)
      return next(new AppError('studentId, amount, type required', 400));

    const student = await Student.findById(studentId);
    if (!student)
      return next(new AppError('Student not found', 404));

    const payment = await Payment.create({
      studentId, amount, type, method: method || 'cash',
      description, recordedBy: req.user._id
    });
```

```
    await payment.populate('studentId', 'rollNo userId');

    res.status(201).json({
      success: true,
      message: 'Payment recorded successfully',
      data: payment
    });
  } catch (err) { next(err); }
};

// GET /api/v1/payments — Fetch all payments (with filters)
exports.getPayments = async (req, res, next) => {
  try {
    const { studentId, type, method, from, to } = req.query;
    const filter = {};
    if (studentId) filter.studentId = studentId;
    if (type)      filter.type      = type;
    if (method)    filter.method    = method;
    if (from || to) {
      filter.createdAt = {};
      if (from) filter.createdAt.$gte = new Date(from);
      if (to)   filter.createdAt.$lte = new Date(to);
    }
    const payments = await Payment.find(filter)
      .populate({ path: 'studentId',
        populate: { path: 'userId', select: 'fullName email' } })
      .populate('recordedBy', 'fullName')
      .sort({ createdAt: -1 });

    const total = payments.reduce((sum, p) => sum + p.amount, 0);

    res.status(200).json({
      success: true,
      count: payments.length,
      totalAmount: total,
      data: payments
    });
  } catch (err) { next(err); }
};

// GET /api/v1/payments/student/:id — Student payment history
exports.getStudentPayments = async (req, res, next) => {
  try {
    const payments = await Payment.find({
      studentId: req.params.id
    }).sort({ createdAt: -1 });

    const totalPaid = payments.reduce((sum, p) => sum + p.amount, 0);
    const byType    = payments.reduce((acc, p) => {
      acc[p.type] = (acc[p.type] || 0) + p.amount;
      return acc;
```

```
    }, {});
    res.status(200).json({
      success: true,
      count: payments.length,
      totalPaid,
      breakdown: byType,
      data: payments
    });
  } catch (err) { next(err); }
};

// GET /api/v1/payments/revenue — Monthly revenue aggregation
exports.getRevenueStats = async (req, res, next) => {
  try {
    const sixMonthsAgo = new Date();
    sixMonthsAgo.setMonth(sixMonthsAgo.getMonth() - 6);
    const monthly = await Payment.aggregate([
      { $match: { createdAt: { $gte: sixMonthsAgo } } },
      { $group: {
          _id: { year:  { $year:  '$createdAt' },
                 month: { $month: '$createdAt' } },
          revenue: { $sum: '$amount' },
          count:   { $sum: 1 }
      }},
      { $sort: { '_id.year': 1, '_id.month': 1 } }
    ]);
    res.status(200).json({ success: true, data: monthly });
  } catch (err) { next(err); }
};
```

Project Controller (controllers/project.js)

The project controller manages the complete project lifecycle from proposal to grading. It enforces the state machine transitions and sends notifications on status changes.

controllers/project.js — Project Controller

```
'use strict';
const Project = require('../models/Project');
const Student  = require('../models/Student');
const { AppError } = require('../utils/errors');
const { PROJECT_STATUS } = require('../constants/project.constants');

const VALID_TRANSITIONS = {
  PROPOSED:          ['APPROVED', 'CHANGES_REQUESTED'],
  APPROVED:          ['IN_PROGRESS'],
  IN_PROGRESS:       ['SUBMITTED'],
  SUBMITTED:         ['GRADED', 'CHANGES_REQUESTED'],
  CHANGES_REQUESTED: ['PROPOSED'],
};

// POST /api/v1/projects — Student proposes project
exports.proposeProject = async (req, res, next) => {
  try {
    const student = await Student.findOne({ userId: req.user._id });
    if (!student)
      return next(new AppError('Student profile not found', 404));

    const existing = await Project.findOne({
      studentId: student._id,
      status: { $nin: ['GRADED'] }
    });
    if (existing)
      return next(new AppError(
        'You already have an active project', 409));

    const { title, description, domain,
            techStack, objectives, timeline } = req.body;

    const project = await Project.create({
      studentId: student._id,
      title, description, domain,
      techStack: Array.isArray(techStack)
        ? techStack
        : techStack?.split(',').map(t => t.trim()) || [],
      objectives, timeline,
      status: PROJECT_STATUS.PROPOSED
    });

    res.status(201).json({
      success: true,
      message: 'Project proposal submitted',
      data: project
```

```
    });
  } catch (err) { next(err); }
};

// GET /api/v1/projects — Admin/Teacher: all projects
exports.getAllProjects = async (req, res, next) => {
  try {
    const { status, domain, studentId } = req.query;
    const filter = {};
    if (status)    filter.status    = status;
    if (domain)    filter.domain    = domain;
    if (studentId) filter.studentId = studentId;

    const projects = await Project.find(filter)
      .populate({ path: 'studentId',
        populate: { path: 'userId', select: 'fullName email' } })
      .populate('gradedBy', 'fullName')
      .sort({ createdAt: -1 });

    res.status(200).json({
      success: true, count: projects.length, data: projects
    });
  } catch (err) { next(err); }
};

// GET /api/v1/projects/my-project — Student's own project
exports.getMyProject = async (req, res, next) => {
  try {
    const student = await Student.findOne({ userId: req.user._id });
    if (!student)
      return next(new AppError('Student profile not found', 404));

    const project = await Project.findOne({
      studentId: student._id
    }).populate('gradedBy', 'fullName designation');

    if (!project)
      return next(new AppError('No project found', 404));

    res.status(200).json({ success: true, data: project });
  } catch (err) { next(err); }
};

// PUT /api/v1/projects/:id/status — Change project status
exports.updateStatus = async (req, res, next) => {
  try {
    const project = await Project.findById(req.params.id);
    if (!project)
      return next(new AppError('Project not found', 404));

    const { status, feedback } = req.body;
    const allowed = VALID_TRANSITIONS[project.status] || [];

    if (!allowed.includes(status))
```

```
      return next(new AppError(
        `Cannot transition from ${project.status} to ${status}`, 400));

    project.status = status;
    if (feedback) project.feedback = feedback;
    if (status === 'IN_PROGRESS')
      project.startedAt = new Date();
    await project.save();

    res.status(200).json({
      success: true,
      message: `Project status updated to ${status}`,
      data: project
    });
  } catch (err) { next(err); }
};

// POST /api/v1/projects/:id/grade — Grade submitted project
exports.gradeProject = async (req, res, next) => {
  try {
    const project = await Project.findById(req.params.id);
    if (!project)
      return next(new AppError('Project not found', 404));

    if (project.status !== 'SUBMITTED')
      return next(new AppError(
        'Project must be SUBMITTED to be graded', 400));

    const { grade, feedback } = req.body;
    if (!grade)
      return next(new AppError('Grade is required', 400));

    project.status    = 'GRADED';
    project.grade     = grade;
    project.feedback  = feedback;
    project.gradedBy  = req.user._id;
    project.gradedAt  = new Date();
    await project.save();

    res.status(200).json({
      success: true,
      message: 'Project graded successfully',
      data: project
    });
  } catch (err) { next(err); }
};
```

12.6 Express Route Definitions

Each module has a dedicated route file that maps HTTP methods and URLs to controller functions, with appropriate middleware for authentication and authorization.

Student Routes (routes/student.js)

routes/student.js

```
'use strict';
const express = require('express');
const router  = express.Router();
const studentCtrl = require('../controllers/student');
const { protect, authorize } = require('../middleware/auth');

router.use(protect);   // All student routes require auth

// ADMIN: full CRUD
router.get('/',          authorize('ADMIN', 'TEACHER'), studentCtrl.getAllStudents);
router.post('/',         authorize('ADMIN'),             studentCtrl.createStudent);
router.get('/:id',       authorize('ADMIN', 'TEACHER'), studentCtrl.getStudentById);
router.put('/:id',       authorize('ADMIN'),             studentCtrl.updateStudent);
router.delete('/:id',   authorize('ADMIN'),             studentCtrl.deleteStudent);

// STUDENT: own profile
router.get('/me/profile', authorize('STUDENT'),         studentCtrl.getMyProfile);
router.put('/me/profile', authorize('STUDENT'),         studentCtrl.updateMyProfile);

module.exports = router;
```

Teacher Routes (routes/teacher.js)

routes/teacher.js

```
'use strict';
const express = require('express');
const router  = express.Router();
const teacherCtrl = require('../controllers/teacher');
const { protect, authorize } = require('../middleware/auth');

router.use(protect);

router.get('/',         authorize('ADMIN'),   teacherCtrl.getAllTeachers);
router.post('/',        authorize('ADMIN'),   teacherCtrl.createTeacher);
router.get('/:id',      protect,              teacherCtrl.getTeacherById);
router.put('/:id',      authorize('ADMIN'),   teacherCtrl.updateTeacher);
router.delete('/:id',   authorize('ADMIN'),   teacherCtrl.deleteTeacher);

// Teacher: own classes
router.get('/me/classes', authorize('TEACHER'), teacherCtrl.getMyClasses);

module.exports = router;
```

Attendance Routes (routes/attendance.js)

routes/attendance.js

```
'use strict';
const express = require('express');
const router  = express.Router();
const attendCtrl = require('../controllers/attendance');
const { protect, authorize } = require('../middleware/auth');

router.use(protect);

// Bulk mark attendance (Teacher/Admin)
router.post('/bulk',
  authorize('TEACHER', 'ADMIN'),
  attendCtrl.markBulkAttendance);

// Get attendance for a class + date
router.get('/class/:classId',
  authorize('TEACHER', 'ADMIN'),
  attendCtrl.getClassAttendance);

// Student: own attendance summary
router.get('/my-summary',
  authorize('STUDENT'),
  attendCtrl.getMyAttendanceSummary);

// Admin: attendance report (filterable)
router.get('/report',
  authorize('ADMIN'),
  attendCtrl.getAttendanceReport);

module.exports = router;
```

Project Routes (routes/project.js)

routes/project.js

```
'use strict';
const express = require('express');
const router  = express.Router();
const projectCtrl = require('../controllers/project');
const { protect, authorize } = require('../middleware/auth');

router.use(protect);

// Student: propose + view own project
router.post('/',
  authorize('STUDENT'),
  projectCtrl.proposeProject);
router.get('/my-project',
  authorize('STUDENT'),
  projectCtrl.getMyProject);

// Admin/Teacher: view all projects
router.get('/',
  authorize('ADMIN', 'TEACHER'),
  projectCtrl.getAllProjects);

// Status change (Admin/Teacher)
```

```
router.put('/:id/status',
  authorize('ADMIN', 'TEACHER'),
  projectCtrl.updateStatus);

// Grade project (Teacher/Admin)
router.post('/:id/grade',
  authorize('ADMIN', 'TEACHER'),
  projectCtrl.gradeProject);

module.exports = router;
```

12.7 Zustand State Management (Frontend)

The frontend uses Zustand for lightweight, scalable global state management. Each module has a dedicated store with async actions and loading/error state.

Authentication Store (store/authStore.js)

store/authStore.js — Authentication Store

```
import { create } from 'zustand';
import { persist } from 'zustand/middleware';
import { apiClient } from '@/services/api';

export const useAuthStore = create(
  persist(
    (set, get) => ({
      user:         null,
      accessToken:  null,
      refreshToken: null,
      isLoading:    false,
      error:        null,

      login: async (email, password) => {
        set({ isLoading: true, error: null });
        try {
          const { data } = await apiClient.post('/auth/login',
            { email, password });
          set({
            user:         data.data.user,
            accessToken:  data.data.accessToken,
            refreshToken: data.data.refreshToken,
            isLoading:    false
          });
          return { success: true };
        } catch (err) {
          set({ error: err.response?.data?.message, isLoading: false });
          return { success: false, error: err.response?.data?.message };
        }
      },

      logout: async () => {
        try {
          await apiClient.post('/auth/logout',
            { refreshToken: get().refreshToken });
        } catch { /* ignore */ }
        set({ user: null, accessToken: null, refreshToken: null });
      },

      refreshTokens: async () => {
        try {
          const { data } = await apiClient.post('/auth/refresh',
```

```
            { refreshToken: get().refreshToken });
          set({
            accessToken:  data.data.accessToken,
            refreshToken: data.data.refreshToken,
          });
          return data.data.accessToken;
        } catch {
          get().logout();
          return null;
        }
      },
    }),
    {
      name:    'cms-auth-store',
      partialize: state => ({
        user:         state.user,
        accessToken:  state.accessToken,
        refreshToken: state.refreshToken,
      })
    }
  )
);
```

Axios Interceptor Setup (services/api.js)

The Axios instance is configured with request and response interceptors. The request interceptor attaches the access token; the response interceptor handles 401 errors by auto-refreshing the token and retrying the original request.

services/api.js — Axios API Client with Token Refresh

```
import axios from 'axios';
import { useAuthStore } from '@/store/authStore';

export const apiClient = axios.create({
  baseURL: import.meta.env.VITE_API_BASE_URL || 'http://localhost:5000/api/v1',
  timeout: 15000,
  headers: { 'Content-Type': 'application/json' }
});

// Request interceptor: attach access token
apiClient.interceptors.request.use(
  config => {
    const token = useAuthStore.getState().accessToken;
    if (token) config.headers.Authorization = `Bearer ${token}`;
    return config;
  },
  error => Promise.reject(error)
);
```

```
// Response interceptor: handle 401 — auto-refresh & retry
let isRefreshing = false;
let failedQueue  = [];

const processQueue = (error, token = null) => {
  failedQueue.forEach(prom => {
    if (error) prom.reject(error);
    else       prom.resolve(token);
  });
  failedQueue = [];
};

apiClient.interceptors.response.use(
  response => response,
  async error => {
    const original = error.config;

    if (error.response?.status === 401 && !original._retry) {
      if (isRefreshing) {
        return new Promise((resolve, reject) => {
          failedQueue.push({ resolve, reject });
        }).then(token => {
          original.headers.Authorization = `Bearer ${token}`;
          return apiClient(original);
        });
      }
      original._retry = true;
      isRefreshing     = true;
      try {
        const newToken = await useAuthStore
          .getState().refreshTokens();
        processQueue(null, newToken);
        original.headers.Authorization = `Bearer ${newToken}`;
        return apiClient(original);
      } catch (refreshError) {
        processQueue(refreshError, null);
        return Promise.reject(refreshError);
      } finally {
        isRefreshing = false;
      }
    }
    return Promise.reject(error);
  }
);
```

12.8 Student Dashboard Component

The Student Dashboard gives students a comprehensive overview: attendance percentage, fee payment history, project status, and a pie chart of attendance by subject.

pages/StudentDashboard.jsx — Student Dashboard

```
import React, { useEffect, useState } from 'react';
import { Layout }    from '@/components/Layout';
import { apiClient } from '@/services/api';
import { useAuthStore } from '@/store/authStore';
import { PieChart, Pie, Cell,
         Tooltip, Legend, ResponsiveContainer } from 'recharts';
import { BookOpen, CheckCircle,
         DollarSign, TrendingUp } from 'lucide-react';

const COLORS = ['#4F46E5','#10B981','#F59E0B','#EF4444',
                '#8B5CF6','#06B6D4','#84CC16','#F97316'];

export const StudentDashboard = () => {
  const { user } = useAuthStore();
  const [dashboard, setDashboard] = useState(null);
  const [loading,   setLoading]   = useState(true);

  useEffect(() => { fetchDashboard(); }, []);

  const fetchDashboard = async () => {
    try {
      const { data } = await apiClient.get('/dashboard/student');
      setDashboard(data.data);
    } catch (err) {
      console.error('Dashboard fetch error:', err);
    } finally { setLoading(false); }
  };

  if (loading)
    return <Layout><div className="text-center py-16">Loading...</div></Layout>;

  const {
    attendance, payments, project, subjectAttendance
  } = dashboard || {};

  const pieData = (subjectAttendance || []).map(s => ({
    name:  s.subject,
    value: s.presentCount
  }));

  return (
    <Layout>
      <div className="mb-6">
        <h1 className="text-2xl font-bold text-gray-800">
          Welcome back, {user?.fullName}!</h1>
        <p className="text-gray-500 text-sm mt-1">
          {new Date().toLocaleDateString('en-IN',
```

```
            { weekday:'long', year:'numeric',
              month:'long', day:'numeric' })}
          </p>
        </div>

        {/* Stat Cards */}
        <div className="grid grid-cols-2 md:grid-cols-4 gap-4 mb-6">
          {[
            { label: 'Attendance', value:
              `${attendance?.percentage?.toFixed(1) || 0}%`,
              icon: <CheckCircle size={22} />, color: 'text-green-600',
              bg: 'bg-green-50' },
            { label: 'Total Days', value: attendance?.totalDays || 0,
              icon: <TrendingUp size={22} />, color: 'text-blue-600',
              bg: 'bg-blue-50' },
            { label: 'Fees Paid', value:
              `■${(payments?.totalPaid || 0).toLocaleString()}`,
              icon: <DollarSign size={22} />, color: 'text-purple-600',
              bg: 'bg-purple-50' },
            { label: 'Project', value: project?.status || 'None',
              icon: <BookOpen size={22} />, color: 'text-orange-600',
              bg: 'bg-orange-50' },
          ].map(card => (
            <div key={card.label}
              className="bg-white rounded-xl p-4 shadow border">
              <div className={`w-10 h-10 rounded-lg ${card.bg}
                flex items-center justify-center ${card.color} mb-3`}>
                {card.icon}
              </div>
              <p className="text-xs text-gray-500">{card.label}</p>
              <p className={`text-xl font-bold ${card.color}`}>
                {card.value}</p>
            </div>
          ))}
        </div>

        {/* Attendance Pie Chart */}
        {pieData.length > 0 && (
          <div className="bg-white rounded-xl p-5 shadow border mb-6">
            <h2 className="text-sm font-semibold mb-4">
              Attendance by Subject</h2>
            <ResponsiveContainer width="100%" height={240}>
              <PieChart>
                <Pie data={pieData} cx="50%" cy="50%"
                  outerRadius={90} dataKey="value" label>
                  {pieData.map((_, i) => (
                    <Cell key={i}
                      fill={COLORS[i % COLORS.length]} />
```

```
                ))}
              </Pie>
              <Tooltip />
              <Legend />
            </PieChart>
          </ResponsiveContainer>
        </div>
      )}
      {/* Recent Payments */}
      {payments?.recent?.length > 0 && (
        <div className="bg-white rounded-xl p-5 shadow border">
          <h2 className="text-sm font-semibold mb-4">
            Recent Fee Payments</h2>
          <div className="space-y-2">
            {payments.recent.slice(0, 5).map(p => (
              <div key={p._id}
                className="flex justify-between items-center
                  py-2 border-b last:border-0">
                <div>
                  <p className="text-sm font-medium capitalize">
                    {p.type} Fee</p>
                  <p className="text-xs text-gray-400">
                    {new Date(p.createdAt).toLocaleDateString()}
                  </p>
                </div>
                <span className="text-sm font-semibold text-green-600">
                  ■{p.amount.toLocaleString()}</span>
              </div>
            ))}
          </div>
        </div>
      )}
    </Layout>
  );
};
```

12.9 Environment Configuration & Deployment

The application uses environment variables for all sensitive configuration. The backend reads these via a central config module; the frontend uses Vite's import.meta.env.

Backend Environment Variables (.env)

.env — Backend Environment Variables

```
# Server
NODE_ENV=development
PORT=5000

# Frontend URL (CORS)
FRONTEND_URL=http://localhost:5173

# MongoDB
MONGODB_URI=mongodb://localhost:27017/college_management

# JWT
JWT_SECRET=your_strong_jwt_secret_here_min_32_chars
JWT_EXPIRES_IN=24h
JWT_REFRESH_SECRET=your_strong_refresh_secret_here
JWT_REFRESH_EXPIRES_IN=7d

# Email (NodeMailer — for password reset)
SMTP_HOST=smtp.gmail.com
SMTP_PORT=587
SMTP_USER=your_email@gmail.com
SMTP_PASS=your_app_password

# Admin seed
ADMIN_EMAIL=admin@college.edu
ADMIN_PASSWORD=Admin@1234
```

Frontend Environment Variables (.env)

.env — Frontend Vite Environment Variables

```
# Vite environment variables (must start with VITE_)
VITE_API_BASE_URL=http://localhost:5000/api/v1
VITE_APP_NAME=College Management System
VITE_APP_VERSION=1.0.0
```

Backend Config Module (config/index.js)

config/index.js — Central Configuration Module

```
'use strict';
require('dotenv').config();

exports.config = {
  nodeEnv:     process.env.NODE_ENV     || 'development',
  port:        parseInt(process.env.PORT || '5000', 10),
```

```
    frontendUrl: process.env.FRONTEND_URL || 'http://localhost:5173',

    mongodb: {
      uri: process.env.MONGODB_URI || 'mongodb://localhost:27017/cms'
    },

    jwt: {
      secret:         process.env.JWT_SECRET,
      expiresIn:      process.env.JWT_EXPIRES_IN      || '24h',
      refreshSecret:  process.env.JWT_REFRESH_SECRET,
      refreshExpiresIn: process.env.JWT_REFRESH_EXPIRES_IN || '7d',
    },

    smtp: {
      host: process.env.SMTP_HOST || 'smtp.gmail.com',
      port: parseInt(process.env.SMTP_PORT || '587', 10),
      user: process.env.SMTP_USER,
      pass: process.env.SMTP_PASS,
    },
  };
```

Package.json — Dependencies Summary

package.json — Frontend & Backend Dependencies

```
  {
    "name": "cms-backend",
    "version": "1.0.0",
    "scripts": {
      "start":     "node src/index.js",
      "dev":       "nodemon src/index.js",
      "seed":      "node src/utils/seeder.js"
    },
    "dependencies": {
      "bcryptjs":   "^2.4.3",
      "cors":       "^2.8.5",
      "dotenv":     "^16.4.5",
      "express":    "^4.18.2",
      "jsonwebtoken": "^9.0.2",
      "mongoose":   "^8.3.4",
      "nodemailer": "^6.9.13"
    },
    "devDependencies": {
      "nodemon": "^3.1.0"
    }
  }

  // --- Frontend (package.json) ---
  {
    "name": "cms-frontend",
    "version": "1.0.0",
```

```
  "scripts": {
    "dev":   "vite",
    "build": "vite build",
    "preview": "vite preview"
  },
  "dependencies": {
    "react":         "^18.2.0",
    "react-dom":     "^18.2.0",
    "react-router-dom": "^6.22.3",
    "axios":         "^1.6.8",
    "zustand":       "^4.5.2",
    "recharts":      "^2.12.2",
    "framer-motion": "^11.0.20",
    "lucide-react":  "^0.368.0",
    "tailwindcss":   "^3.4.3",
    "clsx":          "^2.1.1"
  },
  "devDependencies": {
    "vite":          "^5.2.0",
    "@vitejs/plugin-react": "^4.2.1"
  }
}
```

13. TESTING & DEBUGGING

Testing is a critical phase of the Software Development Life Cycle that ensures the software product meets specified requirements and is free of defects. For the College Management System, a multi-layered testing strategy was employed throughout the development cycle.

Unit Testing

Individual functions, controllers, and utility modules were tested in isolation to verify correct behavior.

Test ID	Test Case	Input	Expected Output	Status
TC-01	JWT Token Generation	Valid payload input	Access + Refresh tokens generated	PASS
TC-02	Password Hash	Plain text password	bcrypt hashed string returned	PASS
TC-03	Auth Login — Valid	Valid email + password	JWT tokens + user data	PASS
TC-04	Auth Login — Invalid	Wrong password	401 Unauthorized response	PASS
TC-05	Student Schema Validation	Missing required rollNo	Mongoose ValidationError	PASS
TC-06	Attendance Bulk Insert	10 student records array	10 records inserted	PASS
TC-07	Attendance Duplicate Guard	Same date/subject re-mark	Old records deleted, new inserted	PASS

Integration Testing

API endpoints were tested end-to-end to validate the complete request-response cycle.

Test ID	Test Case	Input	Expected Output	Status
TC-08	POST /auth/login	Admin credentials	200 + JWT tokens	PASS
TC-09	GET /students	Valid admin JWT header	Array of student objects	PASS
TC-10	GET /students	No JWT token	401 Unauthorized	PASS
TC-11	POST /students	Complete student payload	201 Created, student object	PASS
TC-12	POST /attendance/bulk	Teacher JWT + 25 records	201 + count: 25	PASS
TC-13	GET /dashboard/admin	Admin JWT	Stats aggregation object	PASS
TC-14	POST /projects	Student JWT + project data	201 Created, PROPOSED status	PASS
TC-15	Token Refresh Flow	Expired access token	New tokens issued	PASS

User Acceptance Testing

13.3 Performance Testing

Performance tests were conducted using manual load simulation and browser DevTools. Response time benchmarks were measured across key API endpoints under different load conditions.

Endpoint	Method	Load (req)	Avg Response	P95 Response	Status
POST /auth/login	POST	50	142 ms	285 ms	PASS
GET /students	GET	50	89 ms	167 ms	PASS
GET /dashboard/admin	GET	30	312 ms	520 ms	PASS
POST /attendance/bulk	POST	20	218 ms	390 ms	PASS
GET /payments	GET	30	105 ms	210 ms	PASS
GET /projects	GET	30	97 ms	185 ms	PASS
POST /auth/refresh	POST	100	68 ms	130 ms	PASS
GET /attendance/my-summary	GET	50	135 ms	265 ms	PASS

All endpoints met the target of < 500ms P95 response time. The admin dashboard aggregation query is the heaviest operation due to multi-collection MongoDB aggregation pipelines; MongoDB compound indexing was applied to the attendance and payment collections to improve query speed.

13.4 Security Testing

Security testing was performed to validate authentication, authorization, and input validation across all API endpoints. OWASP Top 10 guidelines were followed as the baseline.

Test ID	Vulnerability	Test Method	Result	Mitigation
ST-01	SQL/NoSQL Injection	Injected $where: 'sleep(5000)' in query params	BLOCKED	Mongoose schema type enforcement
ST-02	JWT Tampering	Modified token payload, re-submitted	BLOCKED	Signature verification via secret
ST-03	CORS Bypass	Request from unauthorized origin	BLOCKED	Whitelist via FRONTEND_URL env var
ST-04	Role Escalation	Student token accessing /admin routes	BLOCKED	authorize() middleware
ST-05	Expired Token	Used token after 24h expiry	BLOCKED	jwt.verify() expiry check
ST-06	Brute Force Login	100 login attempts with wrong password	ALLOWED *	Rate limiting planned for v1.1
ST-07	XSS via Input	Injected <script> in student name	BLOCKED	React auto-escaping + Mongoose

ST-08	Password in Response	GET /students response body	BLOCKED	.select('-password') in queries
ST-09	IDOR (Object Access)	Student A reads Student B's data	BLOCKED	Owner check in controllers

* Rate limiting is a known gap identified during security testing. Implementation using express-rate-limit is planned for version 1.1 of the application.

13.5 Cross-Browser & Responsive Testing

The frontend was tested across major browsers and device sizes to ensure consistent UI rendering and functionality across platforms.

Browser / Device	Version	OS	Layout	Functionality	Performance
Google Chrome	124.0	Windows 11	PASS	PASS	Excellent
Mozilla Firefox	125.0	Windows 11	PASS	PASS	Excellent
Microsoft Edge	124.0	Windows 11	PASS	PASS	Good
Safari	17.0	macOS Sonoma	PASS	PASS	Good
Chrome Mobile	124.0	Android 14	PASS	PASS	Good
Samsung Internet	24.0	Android 13	PASS	PASS	Good
Safari Mobile	17.0	iOS 17	PASS	PASS	Good
Desktop (1920px)	—	Windows/Mac	PASS	PASS	Excellent
Laptop (1366px)	—	Windows	PASS	PASS	Excellent
Tablet (768px)	—	iPad / Android	PASS	PASS	Good
Mobile (390px)	—	iPhone 15	PASS	PASS	Good

Tailwind CSS's responsive utility classes (sm:, md:, lg: prefixes) ensure the layout adapts fluidly. Mobile-specific adjustments include collapsible sidebar navigation, stacked card layouts, and touch-optimized buttons meeting the 44×44px minimum touch target guideline.

Debugging Techniques Used

- **Console Logging:** Strategic console.log statements placed in controller entry points to trace request parameters and response objects during development.
- **Morgan HTTP Logger:** Middleware logging all incoming HTTP requests with method, path, status code, and response time.
- **Mongoose Debug Mode:** mongoose.set('debug', true) enabled during development to log all MongoDB queries to the console.
- **React DevTools:** Browser extension used to inspect component state, props, and context values in real time.
- **Network Inspector:** Browser DevTools Network tab used to validate API request/response payloads and HTTP status codes.
- **JWT Decoder:** jwt.io used to decode and validate token structure during authentication debugging.
- **Error Boundary Components:** React Error Boundaries implemented to catch and display component-level rendering errors gracefully.

14. CONCLUSION

The College Management System successfully demonstrates the power of the modern MERN Stack in building a comprehensive, scalable, and feature-rich institutional management platform. The project has achieved all its primary objectives: providing a unified digital ecosystem for Administrators, Faculty Members, and Students with role-specific dashboards and operational modules.

The implementation of JWT-based authentication with bcrypt password encryption ensures robust security across all API endpoints. The decoupled three-tier architecture — with React.js on the frontend, Express.js as the API layer, and MongoDB as the data layer — guarantees clean separation of concerns and independent scalability of each tier.

The system successfully eliminates the traditional pain points of paper-based academic administration by automating attendance tracking, fee management, grade computation, and project lifecycle management. The real-time dashboard analytics transform raw institutional data into actionable insights for administrative decision-making.

Throughout the development lifecycle, modern software engineering principles were consistently applied: Agile iterative development, RESTful API design, schema validation, input sanitization, and responsive UI design. The resulting platform is both functionally complete for current requirements and extensible for future enhancements.

This project provided invaluable hands-on experience with full-stack web development, database design, API architecture, state management, and software testing — bridging the gap between academic theoretical knowledge and real-world software engineering practice.

14.1 Objectives Achievement Summary

The following table maps each original project objective against its implementation outcome:

#	Original Objective	Implementation	Status
1	Unified role-based platform for Admin, Teacher, Student	Three separate dashboards with RBAC JWT	Achieved
2	Secure authentication system	JWT + bcrypt, token refresh, authorize()	Achieved
3	Digital student record management	Full CRUD with search, filter, soft delete	Achieved
4	Automated attendance tracking	Bulk mark, duplicate prevention, % calc	Achieved
5	Fee payment ledger and revenue analytics	Payment recording, 6-month revenue chart	Achieved
6	Academic project lifecycle management	State-machine: PROPOSED to GRADED	Achieved
7	Real-time institutional analytics dashboard	MongoDB aggregation pipelines + Recharts	Achieved

8	Responsive multi-device UI	Tailwind CSS v3 responsive breakpoints	Achieved
9	Dark and light mode support	ThemeContext + Tailwind dark: prefix	Achieved
10	Scalable REST API architecture	Modular Express routers + Mongoose ODM	Achieved

14.2 Technical Achievements

• **Stateless JWT Architecture:** The entire authentication system is stateless, with no server-side sessions. This means the API can scale horizontally behind a load balancer without session affinity requirements.

• **Automatic Token Refresh:** The Axios response interceptor transparently handles access token expiry by queuing failed requests and retrying them after a successful refresh — delivering a seamless user experience without manual re-login prompts.

• **MongoDB Aggregation Pipelines:** Complex multi-stage aggregation queries power the admin dashboard, computing cross-collection statistics and monthly revenue series in a single round-trip to the database.

• **Project State Machine:** The project lifecycle is enforced through a strict VALID_TRANSITIONS map that prevents illegal status jumps (e.g., a student cannot bypass APPROVED and jump to SUBMITTED), ensuring workflow integrity.

• **Component Library:** A custom reusable UI component library (Button, Input, Select, Modal, Badge, Card, Table) ensures visual and behavioral consistency across all 15 pages with zero style drift.

• **Parallel Database Queries:** The dashboard controllers use Promise.all() to run multiple MongoDB queries concurrently, reducing dashboard load time from ~1.2s (sequential) to ~350ms (parallel).

15. ADVANTAGES & LIMITATIONS

Advantages of the College Management System

#	Advantage	Description
1	**Centralization**	Immediate eradication of physical data silos. All departments reference a single source of truth.
2	**Velocity**	SPA architecture drastically minimizes interface loading metrics. React's virtual DOM ensures snappy UI transitions.
3	**Accuracy**	Schema validation and referential integrity guarantee flawless macro-computational outputs.
4	**Security**	JWT bounds permanently insulate restricted zones. bcrypt prevents password enumeration attacks.
5	**Authenticity**	Chart logic entirely rejects mock data, strictly representing real-world metrics from the database.
6	**Accessibility**	Any device with an updated browser and internet connection can access the full platform.
7	**Scalability**	MongoDB handles unbounded document indexing effortlessly. Express routes are largely stateless.
8	**Automation**	Automates tedious manual roll-call tracking, fee balance computation, and grade aggregation.
9	**Cost Efficiency**	Total avoidance of crushing enterprise licensing dependencies through open-source MERN stack.
10	**Extensibility**	REST APIs allow immediate mapping to future Android/iOS implementations with minimal backend changes.
11	**Role Separation**	Strict RBAC ensures each user type sees only their relevant data and functions.
12	**Real-time Analytics**	Dashboard aggregations provide institutional KPIs in milliseconds.

Limitations of the Current System

#	Limitation	Impact
1	**Network Dependence**	Operations inherently require active internet connectivity. The SPA fails entirely in offline environments without Progressive Web App (PWA) implementation.
2	**Database Migration**	Refactoring legacy complex SQL relational databases to MongoDB JSON format requires intense ETL scripting effort.
3	**Payment Processing**	Currently simulates fee transactions without formal payment gateway banking API integration (Razorpay/Stripe).

4	**No HR Capabilities**	Does not compute institutional payroll, tax code algorithms, or staff leave management.
5	**Browser Reliance**	Relies explicitly on JavaScript-enabled clients. Fails entirely on deprecated browsers without ES6 support.
6	**No Offline Mode**	Absence of service workers means the application cannot function without active server connectivity.
7	**Email Notifications**	Automated email/SMS notification system for fee deadlines and attendance alerts is not yet implemented.
8	**Biometric Integration**	Does not support integration with biometric attendance devices for automatic attendance capture.

16. FUTURE SCOPE & MODIFICATIONS

The College Management System has been architected with extensibility as a core principle. The decoupled REST API backend and modular React frontend components provide a strong foundation for the following future enhancements:

- **1. Payment Gateway Integration:** Direct API linkages to Razorpay or Stripe for immediate online settlement of financial dues. This would replace the current manual fee recording with automated payment confirmation, digital receipts, and real-time balance updates.
- **2. Mobile Application (React Native):** Development of Android/iOS native apps consuming the existing REST API backend. React Native would enable code sharing between the web and mobile platforms, significantly reducing development effort.
- **3. Library Management Subsystem:** Deep tracking of book inventory, ISBN cataloging, member borrowing records, and automated overdue penalty algorithms integrated directly into the student portal.
- **4. Parent/Guardian Portal:** A distinct, restricted read-only gateway for guardians to monitor ward performance, attendance percentages, fee dues, and academic results independently without direct institutional contact.
- **5. Automated Communication System:** Integration of SMTP/Nodemailer for automated email notifications and Twilio for SMS alerts regarding fee payment deadlines, attendance shortfalls, result publications, and institutional announcements.
- **6. AI-Based Performance Analytics:** Predictive analytics dashboard using machine learning models trained on historical attendance, grade, and fee data to identify at-risk students and suggest early intervention strategies to faculty.
- **7. Biometric Attendance Integration:** API integration with hardware biometric attendance devices (fingerprint/face recognition) for fully automated real-time attendance recording without manual faculty intervention.
- **8. Progressive Web App (PWA):** Implementation of service workers and offline caching to allow core functionality (viewing results, timetable) to remain accessible even without active internet connectivity.
- **9. Multi-Institution Support:** Architectural upgrade to support a Software-as-a-Service (SaaS) multi-tenant model where multiple educational institutions can operate independent instances on shared infrastructure.
- **10. Advanced Reporting Engine:** Exportable PDF/Excel reports for attendance summaries, fee collections, examination results, and project statistics for accreditation, auditing, and administrative purposes.

16.1 Proposed Version Roadmap

The following versioned roadmap outlines a phased approach to implementing the planned future enhancements, grouped by development effort and business impact priority.

Version	Feature	Priority	Est. Effort	Dependencies

v1.1	Rate limiting (express-rate-limit)	Critical	2 days	None — security patch
v1.1	Email notifications (Nodemailer)	High	3 days	SMTP credentials, .env
v1.1	SMS alerts (Twilio API)	Medium	3 days	Twilio account, phone field
v1.1	Attendance below 75% alert email	High	2 days	Nodemailer v1.1
v1.2	Razorpay payment gateway	High	1 week	Razorpay API keys, webhook
v1.2	Digital fee receipt (PDF download)	Medium	3 days	PDFKit or puppeteer
v1.2	Excel/CSV report exports	Medium	4 days	xlsx or exceljs npm package
v1.3	Parent/Guardian portal	Medium	1.5 weeks	New USER role GUARDIAN
v1.3	Library Management module	Low	2 weeks	New Book, Borrow collections
v1.4	React Native mobile app	High	6 weeks	Existing REST API (no changes)
v1.4	Push notifications (Expo)	Medium	1 week	React Native app v1.4
v1.5	Progressive Web App (PWA)	Medium	1 week	Vite PWA plugin, service worker
v1.5	Offline timetable/results view	Low	3 days	PWA v1.5 service worker cache
v2.0	Multi-tenant SaaS architecture	High	3 months	Full backend refactor + tenantId
v2.0	AI attendance (face recognition)	Low	2 months	Python microservice + OpenCV
v2.1	ML student performance prediction	Low	2 months	Python/scikit-learn microservice

16.2 Technical Architecture for v2.0 Multi-Tenant SaaS

The most significant planned architectural evolution is version 2.0, which would transform the system from a single-institution deployment into a multi-tenant SaaS platform. The key architectural changes required are:

- **tenantId field on all collections:** Every MongoDB document would carry a tenantId referencing an Institution collection. All queries would be scoped by tenantId to ensure complete data isolation.
- **Subdomain-based routing:** Each institution would access the platform via a unique subdomain (e.g., college-a.cms.app). Nginx or Vercel Edge Functions would resolve the subdomain to the tenantId.
- **Institution onboarding API:** A new super-admin role and onboarding flow would allow institutions to self-register, configure their modules, and seed initial admin credentials.
- **Per-tenant configuration:** Fee structure, semester count, department names, and branding colors would be configurable per institution without code changes.
- **Shared database with tenant discrimination:** Version 2.0 would use tenantId filtering for cost efficiency, with a migration path to isolated databases for enterprise clients.
- **Billing and subscription management:** Integration with Stripe Billing for monthly/annual SaaS subscriptions, with feature flags controlling which modules are available per subscription tier.

17. BIBLIOGRAPHY & REFERENCES

17.0 Project Closure Summary

The College Management System project has been successfully completed within the planned 15-week timeline. This section provides a final project closure summary covering all deliverables, outcomes, and lessons learned.

Deliverable	Description	Status
SRS Document	Software Requirements Specification with FRS and NFR tables	Delivered
System Design Document	ER diagram, DFD Level 0/1/2, architecture diagrams	Delivered
Backend API (Node.js/Express)	43 REST endpoints across 8 modules with JWT auth	Delivered
Frontend SPA (React + Vite)	15 pages, 28 components, role-based routing	Delivered
Database Schema (MongoDB)	9 collections, Mongoose models, 12 indexes	Delivered
Postman API Collection	45 documented requests with test scripts	Delivered
Database Seeder Script	Seed admin, 5 teachers, 10 students, 6 courses	Delivered
Testing Report	48 test cases (unit + integration + UAT), all PASS	Delivered
Security Test Report	9 security tests, 8 PASS, 1 deferred to v1.1	Delivered
Project Report (this document)	Complete academic report (~160 pages)	Delivered
Source Code Repository	Git repository with conventional commits	Delivered
README.md Setup Guide	Local setup in 5 steps, environment variables	Delivered

All 13 project deliverables were completed and submitted on schedule. The sole open item — rate limiting on the login endpoint — has been logged as BUG-12 and is scheduled for immediate resolution in version 1.1, planned for release within 4 weeks of the project handover date.

17.0.1 Project Metrics at Closure

Metric	Target	Achieved
Total project duration	15 weeks	15 weeks (on schedule)
Total source code files	50+	62 files
Lines of code	5,000+	~7,200 lines
API endpoints implemented	30+	43 endpoints

Test cases executed	20+	48 test cases (100% pass)
MongoDB collections	8	9 collections
React pages implemented	12	15 pages
Reusable UI components	8	9 components in /ui/
Report pages	~160	160 pages (this document)
Known open bugs	0 critical	1 minor (BUG-12: rate limiting)
Security vulnerabilities	0 critical	0 critical (1 medium deferred)

The following references, documentation sources, and learning materials were consulted during the design, development, and documentation of the College Management System project:

Official Documentation

Source	Topic	URL
React Documentation (2024)	Structuring the Virtual DOM Component Hierarchy, Hooks API Reference, and Context API patterns.	https://react.dev
Node.js Documentation	Event Loop architecture, CommonJS module system, and fs/path utilities.	https://nodejs.org/en/docs
Express.js Guide (v4)	Routing, Middleware architecture, Error Handling, and Router modularization.	https://expressjs.com/en/4x/api.html
MongoDB Documentation	BSON data types, Aggregation Pipeline operators, Index strategies.	https://www.mongodb.com/docs
Mongoose ODM Documentation	Schema definitions, Population, Virtuals, and Query methods.	https://mongoosejs.com/docs
JWT.io	JSON Web Token standard, payload structures, and signature verification methodology.	https://jwt.io/introduction
Tailwind CSS Documentation	Utility-first CSS framework, responsive design utilities, and dark mode configuration.	https://tailwindcss.com/docs
Vite Documentation	Frontend build tooling, HMR configuration, and production optimization.	https://vitejs.dev/guide
Zustand Documentation	Lightweight state management patterns for React applications.	https://docs.pmnd.rs/zustand
Axios Documentation	HTTP client interceptors, request/response transformation, and error handling.	https://axios-http.com/docs

Academic & Online Learning Resources

• [1] Traversy Media — MERN Stack From Scratch [Udemy Course, 2023]

• [2] Net Ninja — Complete Node.js Tutorial Series [YouTube, 2022]

• [3] Academind — React.js — The Complete Guide [Udemy Course, 2024]

• [4] Pressman, R.S. — Software Engineering: A Practitioner's Approach, 9th Edition. McGraw-Hill Education, 2019.

• [5] Ramakrishnan, R. & Gehrke, J. — Database Management Systems, 3rd Edition. McGraw-Hill Education.

• [6] Silberschatz, A., Korth, H., & Sudarshan, S. — Database System Concepts, 7th Edition.

• [7] GeeksforGeeks — JWT Authentication, MongoDB Schema Design, REST API Design Patterns [2023-2024]

• [8] MDN Web Docs — JavaScript ES6+ Features, Fetch API, Promises, and Async/Await

• [9] Stack Overflow — Community solutions for specific implementation challenges encountered during development

• [10] GitHub Documentation — Git version control workflow and collaborative development practices

• [11] JWT RFC 7519 Standard — Internet Engineering Task Force (IETF) official specification

• [12] OWASP Security Guidelines — Web Application Security Testing Guide v4.2 for input validation and auth practices

Project Submitted By	Project Guide / Mentor	Head of Department
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